Negotiated Binding Agreements*

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Abstract

I study binding agreements over play in a game and the possibility of inefficient agreements when mutual confirmation is required. I propose a negotiation protocol where, in each round, agents propose actions from the underlying game that they could agree to, signalling the agreement they would like. The protocol terminates when proposals are mutually confirmed. The model is solved using Subgame Perfect Equilibrium. Since agents only propose actions they could agree to, the agreement outcomes exhibit a self-generating structure. A full characterisation is provided for two-player games, based on individual punishments that must themselves be agreements. These individual punishments are used for sufficiency in n-player games and a necessary iterative rationality constraint is introduced. I show that inefficient agreements persist under discounting and imperfect confirmation. I also extend the concept to coalitional agreements within the negotiation, where generalisations of the main results hold.

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1 Introduction

Negotiated binding agreements over actions, from international trade deals to labour contracts, shape outcomes across much of economic life. Economic theory, following the logic of the Coase Theorem, predicts that when mutual gains exist and there are no frictions, fully rational agents should bargain until efficiency is achieved.[1] Even when agreement among

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[1] Although the statement is usually made for environments with transfers, the idea is applied more broadly. I remain agnostic on whether transfers are available within the underlying game. In either case the main results apply and inefficiencies are possible.

all parties is not achievable, existing models often imply that the most efficient feasible outcome is reached among those who do agree (Ray and Vohra, 1997; Ellingsen and Paltseva, 2016). Yet inefficient agreements are widespread. Trade agreements maintain positive tariffs despite mutual gains from liberalisation, climate treaties fall short of optimal emissions targets, and the post-Brexit negotiations between the UK and EU produced a deal that imposed frictions both sides had, from the outset, preferred to avoid.

In practice, such agreements are reached in a particular way, with two recurring features. They bind only by mutual consent: an outcome takes effect only once every party signs off on the same terms, so any one of them can hold it up. Additionally, they are negotiated in good faith. A central feature of good-faith in negotiation is that the terms put forward should represent genuine possibilities for agreement, which those proposing them would be willing to have accepted. But what constitutes a possibility for agreement depends on what the parties could agree to, so the terms put forward must be consistent with the agreements that negotiation could produce.[2]

I study a model of negotiation that captures these two features. Agents propose actions to undertake in an underlying game, and the proposed action profile becomes binding only when it is mutually confirmed. In keeping with good faith, each player may propose only agreement actions: actions that could form part of an agreement under the same equilibrium strategies. This gives the proposals a self-generating structure. Importantly, good faith is a requirement on each player's conduct, not on their ability to coordinate: their individual proposals need not be mutually agreeable. I solve for subgame-perfect equilibria under complete information and, in the baseline, without discounting, isolating the role of mutual consent in determining which agreements can be sustained. Nonetheless, inefficient agreements can persist. Even when a Pareto improvement is available, a player may correctly anticipate that a unilateral proposal towards it would remain unconfirmed, and hence unimplemented. Neither player therefore has an incentive to initiate the change. This raises two questions: which agreements can persist, and how inefficient can they be?

I establish two main results. The first is a full characterisation for two players. A profile is sustained if and only if it gives each player at least a player-specific threshold, the worst agreement that player can credibly be held to. The threshold is pinned down by two requirements on the punishment that enforces an agreement: it must give the punished player a best response in the underlying game, and give the punisher at least what their own threshold guarantees, so that each is willing to carry the other's punishment out.

[2]This conception draws on Iklé (1964), as discussed by Farnsworth (1987), who identifies among the meanings of good faith both wanting an agreement of the kind being negotiated and being eager to have one's terms accepted. Similarly, the UNIDROIT Principles treat negotiating without an intention of reaching agreement as bad faith (International Institute for the Unification of Private Law (UNIDROIT), 2016), while United States labour law requires a sincere effort to reach agreement.

These thresholds are built from the underlying game, in the manner of the self-enforcing punishments of repeated games (Fudenberg and Maskin, 1986; Abreu et al., 1994) and the commitment folk theorems of contract theory (Tennenholtz, 2004; Kalai et al., 2010; Peters and Szentes, 2012). What distinguishes the present thresholds is the discipline good faith imposes: a punishment cannot be an arbitrary threat, but must itself be an agreement that each player would consent to. This rules out threats that are not *contemporaneously* credible,[3] so the characterisation is more restrictive than a folk theorem and places player-specific payoff bounds on sustainable agreements.

The second treats n-player games, where the two-player characterisation provides sufficient conditions for agreement. To derive necessary conditions, I introduce the concept of "iterated deletion of individually irrational actions" in the underlying game. An action is eliminated if, even under the most optimistic beliefs about others' actions, it yields a lower payoff than some guaranteed response. This process identifies actions that cannot credibly be proposed or agreed upon. Any equilibrium agreement must survive this iterative process. In many games, the necessary and sufficient conditions lead to a tight characterisation on outcomes.

To illustrate these results, I analyse two benchmark games. First, a Cournot duopoly with asymmetric costs shows how bargaining power emerges endogenously and shapes the set of sustainable agreement outcomes. Compared with folk theorem predictions, the set of sustainable agreements excludes some highly inefficient profiles and favours the firm with lower marginal costs. Even under good faith, the admissible proposals can be surprisingly extreme. The low-cost firm can propose to flood the other out of the market altogether, leaving the high-cost firm with zero profit, and the high-cost firm would still confirm it, because resisting would earn it nothing either way. Second, a first-price auction with heterogeneous bidders demonstrates how minimal payoff guarantees constrain the allocation: bidders with low valuations cannot be awarded the good with probability one under any equilibrium agreement.

The baseline deliberately makes delay costless, so inefficiency cannot be attributed to impatience. I show that this conclusion is robust to discounting and imperfect confirmation. In the Cournot example, punishments can be constructed that approach the original ones, allowing agreements strictly above the original payoff floors to remain sustainable when players are sufficiently patient and mistakes sufficiently rare.

Finally, I extend the framework to coalitional deviation, letting players commit not only

[3]The contracting literature imposes no such restriction: a flow of payoffs lets a punishment today be rewarded with prizes tomorrow, which secures enforcement in infinitely repeated games.

to the action they will take but to their conduct in the negotiation itself, in coalitions. I take the coalitions to partition the players and define a generalised equilibrium, the $\mathcal{C}$-Subgame Perfect Equilibrium, under which no coalition can jointly deviate to improve all of its members. The sustainable outcomes refine the β-core of Aumann (1961) through a recursive deletion that justifies the punishments it rests on, and for a partition the characterisation specialises cleanly, connecting to the coalitional equilibrium of Ray and Vohra (1997). Coalitional deviation can restore efficiency in some games and not in others.

This approach complements existing work on bargaining, repeated games, and contract theory by providing a tractable model of agreement formation without relying on a mediator to coordinate outcomes. It identifies a source of inefficiency arising from mutual confirmation, even under complete information and good-faith negotiation, and clarifies the restrictions that these institutional features place on sustainable agreements.

2 Model

The underlying game is a strategic-form game $G = \langle N, (A_i, u_i)_{i \in N} \rangle$, where $N = \{1, \ldots, n\}$ is a finite set of players and A_i is the action set of player i, with typical element a_i. The set of action profiles is $A = \times_{i \in N} A_i$, with typical element $a = (a_i)_{i \in N}$, and each player i has a bounded payoff function $u_i : A \to \mathbb{R}$. Write $A_{-i} = \times_{j \neq i} A_j$.

Two features of this specification are worth noting. First, an agreement binds the players before G is played, so an agreement is a commitment to a profile of actions, and G may equally be the strategic form of a dynamic game, with A_i the set of strategies of player i and an agreement a simultaneous commitment to strategies of that game, or a static game. The results apply in either case. For concreteness the examples throughout are static games, which keeps the notation light. Second, A_i may itself be a space of mixed strategies, $A_i = \Delta(\bar{A}_i)$ for a finite set of base actions $\bar{A}_i$, in which case an agreement is reached over mixed strategies of that base game. This amounts to giving each player its own randomisation device over $\bar{A}_i$ and letting it commit to the device, so the agreed profile induces independent mixing over the base-action profiles. I do not allow agents to randomise over their proposals in the bargaining game, though it is straightforward to extend the analysis to a mixed action space in the underlying game in this way.[4]

The *negotiation game* over G has potentially infinitely many periods. In each period every agent simultaneously proposes an action $a_i \in A_i$ to play in G, and observes the others' proposals. An agent then either confirms the standing profile, by proposing the same action

[4]For some games no pure profile is agreeable under the conditions that follow. In matching pennies, for instance, no pure action profile satisfies them for both players at once, so the game has no Negotiated Binding Agreement in pure actions. Taking the underlying game to be its mixed extension restores one, as the uniform profile, which equalises the players' payoffs, is then agreeable.

again, or makes a new proposal.[5] If all agents confirm, an agreement is reached and the confirmed profile is played in G. Otherwise negotiation continues, and if agreement is never reached the outcome is *perpetual disagreement*.[6] Throughout, an agent may propose only an agreement action, the good-faith restriction made precise below.

Formally, a partial history is a finite sequence $h = (a^1, \dots, a^k)$, where $a^t = (a_i^t)_{i \in N}$ is the profile of proposals made in period t, with no consecutive repetition, $a^t \neq a^{t-1}$ for all $t \leq k$. Write H for the set of all partial histories. Because proposals within a period are made simultaneously, no history records a period in which only some agents have proposed.[7]

A history is terminal if either:

1. the same profile is proposed in two consecutive periods, with no earlier consecutive repetition. That is, $z = (a^1, \dots, a^{k-1}, a^k)$ with $a^k = a^{k-1}$ and $a^m \neq a^{m-1}$ for all $m < k$. The set of such histories is denoted Z', and I call them histories *with agreement*, or

2. proposals continue indefinitely with no profile ever repeated in consecutive periods. The set of such histories is denoted Z'', and I call them histories *with perpetual disagreement*.

Let the set of all terminal histories be given by $Z = Z' \cup Z''$.

Let $U_i : Z \to \mathbb{R}$ be player i's payoff in the negotiation game. At an agreement, each player receives the payoff of the agreed profile: for $z = (a^1, \dots, a^k) \in Z'$, $U_i(z) = u_i(a^k)$ for all $i \in N$.

At perpetual disagreement, the payoff lies between the lim inf and the lim sup of the proposed payoffs $u_i(a^t)$. Formally, for $z = (a^1, a^2, \dots) \in Z''$, I take

$$U_i(z) \in \Big[\liminf_{t \to \infty} u_i(a^t),\ \limsup_{t \to \infty} u_i(a^t) \Big].$$

This specification keeps payoffs pinned down by what is proposed, in aggregate, within the negotiation game, imposes no cost of delay, and gives individual proposals no direct payoff consequences. I discuss it at length below.

A strategy for player i maps each partial history to a proposal, $s_i : H \to A_i$, with $s_i(h)$ the action i proposes after history h. Let S_i be the set of such strategies, and let $s = (s_i)_{i \in N}$ be a strategy profile, so that $s(h) = (s_i(h))_{i \in N}$.

[5] The particular confirmation technology matters little for the analysis and is adopted here for notational convenience.

[6] Formally, this game is similar to that used in the farsighted stable set for games, which is discussed at length in the literature review in section 7.

[7] A previous working paper version considered the extension of non-simultaneous proposals where the main insights are upheld.

For a partial history h and strategy profile s, let $(s|h) \in Z$ be the terminal history reached by following s from h, namely $(s|h) = (h, a^1, a^2, \dots)$ with $a^1 = s(h)$ and $a^t = s(h, a^1, \dots, a^{t-1})$. Let $U_i(s|h)$ be player i's payoff at $(s|h)$, where at a perpetual-disagreement continuation the lim inf and lim sup are taken over the continuation proposals alone. For a terminal history $z = (a^1, \dots, a^k) \in Z'$, write $a(z) = a^k$ for the agreed profile and $a_i(z)$ for i's component, and correspondingly $a(s|h)$ for the agreement reached from h under s.

The negotiation game resembles a repeated game in its partial histories and in the payoff to perpetual disagreement, but it differs in two respects. A repeated game has a single kind of terminal history, in which the underlying game is played a fixed number of times, finite or infinite, whereas the negotiation game has two: histories that reach agreement, which are finite, and histories that do not, which are infinite. And where a repeated game delivers a flow payoff in every period, here a payoff is realised only on termination. Both departures are common across the related literature (Kalai, 1981; Bhaskar, 1989; Kimya, 2020; Nishihara, 2022).

2.1. Discussion of Perpetual Disagreement Payoffs

Recall that the perpetual disagreement payoffs are such that

$$U_i(z) \in [\liminf_{t\to\infty} u_i(a^t), \limsup_{t\to\infty} u_i(a^t)]$$

for $z \in Z''$. This restriction is consistent with a standard probabilistic termination model, where the proposal today is implemented with probability $(1-\delta)$ for each period, while the process continues with probability δ, taking δ to 1. Therefore, this can also be interpreted as a limiting version of the condition used within Kimya (2020), where there is a probability that the negotiation will end at the currently proposed actions.[8] This is formalised by the following lemma, and the proof is provided in the appendix.

Lemma 1. *Let $z = (a^1, a^2, \dots) \in Z''$ and write $V_i(\delta) = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1} u_i(a^t)$. Then $V_i(\delta)$ is bounded, and every sub-sequential limit of $V_i(\delta)$ as $\delta \to 1$ lies in*

$$\Big[\liminf_{k\to\infty} u_i(a^k),\ \limsup_{k\to\infty} u_i(a^k)\Big].$$

In particular, for any sequence $\delta_m \to 1$ along which $V_i(\delta_m)$ converges its limit lies in this interval. Such a sequence always exists, since V_i is bounded, though the limit may depend

[8] Similar notions also exist in the context of Rubinstein (1982) bargaining, where Busch and Wen (1995) take a game to be played in each rejection phase, which is implemented with probability $1-\delta$ and continuation occurs to a new proposal happens with probability δ, allowing for an endogenous outside option.

on the sequence and $V_i(\delta)$ need not converge as $\delta \to 1$.

The value of perpetual disagreement may be taken to be *any* point in this interval, which frees the main results from the details of how confirmation is modelled: all that matters is that agreement requires simultaneous confirmation by every player. The protocol then admits an accept/decline reading, in which a designated action keeps negotiating, re-proposing the standing profile counts as accepting it, and unanimous acceptance makes it binding. If each player accepts in error with vanishing probability, independently of the payoff to no agreement, Lemma 1 places the induced disagreement value in the interval, at a point set by how the termination probability vanishes. This is the accidental-acceptance micro-foundation, which Section 5 makes fully explicit under discounting. A natural alternative, standard in bargaining, would instead fix the payoff to perpetual disagreement at a single value weakly below every payoff of G, normalising payoffs to be positive and disagreement to 0.

Remark 1. *If the payoff to perpetual disagreement is normalised to lie weakly below every payoff of G, and each player has at least two actions, then any profile that all players strictly prefer to disagreement is a subgame perfect equilibrium outcome of the negotiation game.*

A folk theorem therefore returns. To support a profile a^*, agents confirm it on the path and, after any deviation, rotate between two profiles so that none is repeated and perpetual disagreement follows, holding the deviator to the low value. Since every player prefers a^* to disagreement, no deviation pays. The construction is routine, and is given in the appendix.[9]

However, some of these folk theorem equilibria are fragile. Each holds some player below the range of proposed payoffs, which requires sustaining perpetual disagreement, and this cannot be done once confirmation is imperfect: if players confirm the standing profile in error with small, independent probability, agreement is reached in finite time almost surely, so no player can be held below a strictly positive floor, as the discounting analysis of Section 5 confirms. The interval specification instead rules out punishments that rely on holding a player below the range of proposed payoffs. Section 5 examines directly which of the resulting agreements remain sustainable under imperfect confirmation and costly delay.

Taking the disagreement value to range over the whole interval, rather than fixing it, ensures that neither the exact rule by which disagreement payoffs are aggregated nor the precise form of the friction affects the predictions. The same outcomes are equilibria whether perpetual disagreement is valued by the probabilistic-termination model of Kimya (2020),

[9] The good-faith restriction introduced below does not disturb this. A single equilibrium can confirm every profile as an agreement at some history, before settling on the path, so that under it every action is an agreement action, $A_i^* = A_i$, and good faith restricts nothing. The folk theorem therefore holds for Negotiated Binding Agreements as well.

by the limit of means as in Rubinstein (1994) and Aumann and Shapley (1994), or by any other rule that delivers a value in the interval.[10] The two-player characterisation and the n-player conditions hold for any such value. One later coalitional result, noted where it appears, instead fixes the value at the lim inf.

2.2. Solution Concept

As discussed in the Introduction, a recognised interpretation of negotiating in good faith involves both a desire to reach agreement and a willingness to have one's proposed terms accepted. I focus on a natural implication of these ideas: a proposal should represent an action the player would genuinely be willing to undertake as part of an agreement, rather than one that could never be agreed to.

Whether an action can form part of an agreement depends, however, on what the other players are willing to undertake. I therefore require each equilibrium proposal to be an *agreement action*: an action that forms part of an agreement reached under the same equilibrium strategies following some history. I call this *respecting signalling of agreement*. The negotiation protocol is a dynamic game of complete information, so Subgame Perfect Equilibrium (SPE) provides the underlying solution concept. Formally:

Definition. *a^* is a Negotiated Binding Agreement if there exists a strategy profile s^* of the negotiation game such that:*

1. *$a(s^* \mid \emptyset) = a^*$, with $(s^* \mid \emptyset) \in Z'$.*
2. *s^* is a Subgame Perfect Equilibrium.*
3. *s^* respects signalling of agreement: for every $i \in N$ and $h \in H$, $s_i^*(h) \in A_i^*(s^*)$, where*

$$A_i^*(s^*) = \left\{ a_i \in A_i : a_i = a_i(s^* \mid h) \text{ for some } h \in H \text{ with } (s^* \mid h) \in Z' \right\}$$

is the set of player i's actions in agreements reached under s^*.

The third condition ensures that proposals and agreements are mutually consistent. An action qualifies as a genuine possibility for agreement only if, following some history, the players' equilibrium strategies lead to an agreement containing it. In particular, a player cannot justify a proposal by appealing to concessions that the others would never make

[10] There is a further, belief-based reading of why the value is taken over the whole interval, which I give once good faith is in place: the interval is the range of values a player could attach to never agreeing under different expectations about which agreement action of the other the path approaches, so taking all of it is robustness to those expectations as well.

under those same strategies. The restriction therefore requires that the willingness to agree implicit in a proposal is reflected in the players' equilibrium behaviour.

Importantly, good faith is a requirement on each player's individual conduct, not on their ability to coordinate. Each proposed action must form part of some agreement, but the profile currently proposed by all players need not itself be mutually agreeable. Nor must an agreement involving a proposed action be expected at the current history. Players may therefore make incompatible proposals, begin with ambitious demands, and subsequently moderate them. In the Cournot duopoly below, for example, the more efficient firm can propose its monopoly quantity, since that quantity forms part of a sustainable agreement in which the other firm produces nothing.[11]

Because $A_i^*(s^*)$ is determined by the agreements reached under the very strategies whose proposals it restricts, admissibility is endogenous and self-generating, in the spirit of Abreu (1988). The requirement applies at every history, but not to deviations: a player remains free to deviate to any action in A_i.[12] The definition allows miscoordination and delay, which are costless in the baseline. Section 5 considers impatience. Any agreement reached following some history can be shifted to the initial history, so it suffices to characterise the possible agreement outcomes.

The good-faith restriction also provides an interpretation of the perpetual disagreement payoffs in Section 2.1.. If negotiation continues indefinitely, players continue to propose actions that could form part of an agreement under the same equilibrium strategies, even off the equilibrium agreement path. In this sense, the payoff from never reaching agreement is evaluated by reference to the alternative possibilities for agreement that the players continue to put forward. The interval restriction allows for different ways of evaluating the payoffs of these proposals, while ruling out punishments that depend on assigning perpetual disagreement a value outside their long-run range.

Good faith is therefore an additional discipline on equilibrium conduct, not something guaranteed by strategic rationality alone. Without it, proposals may serve as threats even when they have no role in any agreement. The following benchmark illustrates the permissiveness of SPE when proposals are unrestricted.

Remark 2 (Benchmark folk theorem)**.** *For $N = \{1, 2\}$, if proposals are unrestricted and perpetual disagreement is valued at the* $\liminf$ *of proposed payoffs, every profile a with*

[11] A previous working paper version considered proposals of entire action *profiles*, rather than individual actions, with the main insights preserved.

[12] It will become apparent that deviations outside of A_i^* are not relevant, but are included in the spirit of SPE.

$u_i(a) \geq \underline{v}_i$ *for* $i = 1, 2$ *is an SPE agreement outcome, where*

$$\underline{v}_i := \min_{a_{-i}} \max_{a_i} u_i(a_i, a_{-i})$$

is player i*'s minmax value. The* $\liminf$ *permits the strongest punishment and hence the largest such set. Higher valuations of perpetual disagreement, up to the* $\limsup$*, may rule out some of these outcomes. The enforcing threats may involve actions that never occur in an agreement, which the good-faith restriction rules out.*

The benchmark shows why a good-faith requirement can have a substantive institutional role: SPE alone does not rule out threats disconnected from possible agreements, whereas imposing good faith can eliminate outcomes sustained by such threats. This offers a rationale for why legal duties and social norms may discipline negotiation rather than leaving the matter entirely to strategic incentives. Good faith does not, however, guarantee efficiency. In the two-player characterisation, the relevant threshold is the worst agreement to which each player can credibly be held, rather than the unrestricted minmax. Since these punishments must themselves be agreements, the thresholds are determined jointly with the agreements they sustain. The next section characterises them using the underlying game alone.

3 Negotiated Binding Agreement Action Profiles for Two-Player Games

In this section I give a full characterisation of the Negotiated Binding Agreements of an underlying two-player game whose action sets are compact and whose payoffs are continuous. The logic is the one previewed in the introduction. A player will agree to an outcome whenever it is better than the worst agreement that player can be held to, so for each player i there is a worst agreement $\underline{a}^i$, with $u_i(\underline{a}^i) \leq u_i(a)$ for every agreement a. For $\underline{a}^i$ to serve as a credible punishment it must itself be an agreement, which both players must be willing to accept: the punished player must have no profitable unilateral deviation from it in G, and the other must receive at least their own worst-agreement payoff, so that each would rather punish the other than be punished themselves. I develop this through a Cournot duopoly with linear demand and heterogeneous marginal costs, and use it to draw the distinction between Negotiated Binding Agreements, the player-specific punishments of Fudenberg and Maskin (1986); Abreu et al. (1994), and commitment folk theorems (Peters and Szentes, 2012).

3.1. Leading Example and Preview of Results for Two-Player Games

Consider a Cournot duopoly as the underlying game G. Each firm i chooses a quantity $q_i \in A_i = [0, b]$, and inverse demand is $p(q_1, q_2) = \max\{b - q_1 - q_2, 0\}$. Firms have potentially heterogeneous marginal costs c_1 and c_2. Without loss of generality, label them so that $c_1 \geq c_2 \geq 0$, and assume firm 1 is a viable competitor, $\frac{b+c_2}{2} \geq c_1$.[13] Profits are $\pi_i(q_1, q_2) = q_i(p(q_1, q_2) - c_i)$.

The best responses in G are:

$$q_i^*(q_{-i}) = \begin{cases} 0 & \text{if } q_{-i} \geq b - c_i \\ \frac{b - c_i - q_{-i}}{2} & \text{if } q_{-i} < b - c_i \end{cases}$$

The Nash equilibrium of G is $\left(\frac{b+c_2-2c_1}{3}, \frac{b+c_1-2c_2}{3}\right)$, with payoffs $\left(\left(\frac{b+c_2-2c_1}{3}\right)^2, \left(\frac{b+c_1-2c_2}{3}\right)^2\right)$.

Consider a profile (q_1^*, q_2^*) with $\pi_1(q_1^*, q_2^*) \geq 0$ and $\pi_2(q_1^*, q_2^*) \geq (c_1 - c_2)^2$. Such profiles exist, since the viable-competitor assumption $\frac{b+c_2}{2} \geq c_1$ gives $\left(\frac{b-c_2}{2}\right)^2 \geq (c_1 - c_2)^2$. I show that these are precisely the Negotiated Binding Agreement action profiles.

Consider strategies built from two punishment profiles, $\underline{q}^1 = (0,\, b - c_1)$ and $\underline{q}^2 = (b - 2c_1 + c_2,\, c_1 - c_2)$, which punish firm 1 and firm 2 respectively. On the path both firms propose (q_1^*, q_2^*) and confirm it. If a firm deviates at any point of play, they move to that firm's punishment and settles there, so the firms go on to agree on the punishment profile.[14]

The punishments fix the firms' payoffs. Under firm 1's punishment the firms agree on $\underline{q}^1$, so firm 1 earns $\pi_1(\underline{q}^1) = 0$ and firm 2 earns $\pi_2(\underline{q}^1) = (b - c_1)(c_1 - c_2)$. Under firm 2's punishment they agree on $\underline{q}^2$, so firm 1 earns $\pi_1(\underline{q}^2) = 0$ and firm 2 earns $\pi_2(\underline{q}^2) = (c_1 - c_2)^2$. When a firm deviates the other floods the market, but only partially, because the punishment must itself remain an agreement the punisher will accept. Firm 1 is held to zero under either punishment, so it floods firm 2 only up to the point where its own profit would turn negative, and when $c_1 > c_2$ it cannot drive firm 2 out entirely without a loss.

These strategies are a subgame perfect equilibrium, which verifies that q^* is a Negotiated Binding Agreement. In its own punishment a firm faces a fixed opponent quantity and can do no better than best-respond and agree, earning its punishment payoff, so it cannot gain by deviating. On the agreement path, and while carrying out the other's punishment, a

[13] This is exactly the condition for firm 1 to produce a positive quantity in the Cournot Nash equilibrium, so that it is a genuine competitor. At the borderline $c_1 = \frac{b+c_2}{2}$ firm 1's equilibrium quantity falls to zero.

[14] A departing firm must not be able to confirm the profile it departed to. To prevent this, the punisher first spends one period proposing its own component of the punishment, so the departed-to profile cannot be repeated, before the punishment profile itself is settled. The formal construction, with this rerouting period, is given in the proof of Theorem 1.

firm that departs only triggers its own punishment, which pays it no more: on the path firm 1 earns $\pi_1(q^*) \geq 0$ and firm 2 earns $\pi_2(q^*) \geq (c_1 - c_2)^2$, each at least its punishment payoff. The one substantive condition is that firm 2 prefer carrying out firm 1's punishment to provoking its own, which is exactly the viable-competitor assumption $(b - c_1)(c_1 - c_2) \geq (c_1 - c_2)^2$. Because play reaches an agreement from every history, every proposed action is part of some agreement, so signalling of agreement holds.

I now show that any Negotiated Binding Agreement gives each firm at least its punishment payoff $\pi_i(\underline{q}^i)$. Because each firm may propose only agreement actions, every proposal lies in Q^*, the set of quantities arising in some agreement. In any equilibrium firm i can best-respond statically in every period, so in any agreement it earns at least $\inf_{q_{-i} \in Q^*_{-i}} \max_{q_i} \pi_i(q_i, q_{-i})$, the lowest payoff an agreeable opponent quantity can push it to. In this example the infimum is achieved by the punishment profile $\underline{q}^i$ constructed above, which is itself an agreement, so $\pi_i(\underline{q}^i)$ is firm i's worst agreement payoff.[15] The agreement set is also monotone: if a profile is an agreement, so is any profile both firms prefer, since the same punishments enforce it.

It remains to compute the two floors. Firm 1's punishment gives $\pi_1(\underline{q}^1) = 0$, the lowest profit it can be held to, so every agreement gives firm 1 at least 0. Firm 2's floor is reached when firm 1 produces the most it can while still willing to agree, $b - 2c_1 + c_2$ (at which $\pi_1 = 0$), and firm 2 best-responds. This leaves firm 2 with $(c_1 - c_2)^2$. The Negotiated Binding Agreement action profiles are therefore exactly those with $\pi_1 \geq 0$ and $\pi_2 \geq (c_1 - c_2)^2$.

This analysis yields natural comparative statics and a comparison with the player-specific punishments of Fudenberg and Maskin (1986); Abreu et al. (1994) and the commitment folk theorems of Peters and Szentes (2012). When $c_1 = c_2$, the firms can agree to any payoffs above their individually rational level of 0, and the Negotiated Binding Agreement set coincides with the sets delivered by the commitment folk theorem and by individual punishments. When instead $c_1 > c_2$, those two sets are unchanged, but the Negotiated Binding Agreement set shrinks, reflecting firm 2's greater bargaining power: firm 1 will neither propose nor accept an agreement against its own interest. In the extreme case where firm 1 is no longer a viable competitor, $c_1 = \frac{b+c_2}{2}$, firm 2's floor rises to its monopoly profit, $\pi_2(\underline{q}^2) = \left(\frac{b-c_2}{2}\right)^2$.[16] Each firm's bargaining power is therefore dictated by the difference in marginal costs, a parameter of the *underlying game*, rather than by the bargaining protocol. The proposals good faith admits can be surprisingly extreme. The low-cost firm can pro-

[15] In general the infimum need not be attained within the agreeable quantities. The formal proof of Theorem 1 passes to the closure, using compactness of A_{-i} and continuity of π_i to obtain a limiting profile achieving the bound.

[16] If $c_1 > \frac{b+c_2}{2}$ then the only outcome that can be supported by a Negotiated Binding Agreement is $q_1^* = 0$, $q_2^* = \frac{b-c_2}{2}$, while under commitment folk theorems and individual punishments all individually rational payoffs would still be supported.

pose to flood the other out of the market altogether, producing so much that the high-cost firm earns nothing, and this is a Negotiated Binding Agreement. The high-cost firm would confirm it, because rejecting cannot improve its payoff under the supporting equilibrium, where the punishment it would carry out, producing enough to make the low-cost firm weakly worse off, earns it nothing either way. A demand this one-sided is sustained in good faith, because the outcome it names is one the losing firm would still accept. Inefficient agreements can nonetheless arise.

3.2. Results

The logic of the previous example generalises. The threat that enforces an agreement is itself an agreement, the worst one a player can be held to, since a credible punishment must be a profile the players would actually confirm. A profile is sustainable exactly when it gives each player at least this worst-agreement threshold, which the following theorem makes precise.

Theorem 1 (Full Characterisation for Two-Player Games)**.** *Let $N = \{1, 2\}$, and for $i = 1, 2$ let A_i be a compact subset of a metric space and u_i continuous. Then a^* is a Negotiated Binding Agreement action profile if and only if there exist $\underline{a}^1, \underline{a}^2 \in A$ such that, for $i, j \in \{1, 2\}$ with $i \neq j$,*

1. $\underline{a}^i_i \in \arg\max_{a_i \in A_i} u_i(a_i, \underline{a}^i_{-i})$,
2. $u_i(\underline{a}^j) \geq u_i(\underline{a}^i)$,
3. $u_i(a^*) \geq u_i(\underline{a}^i)$.

Any pure Nash equilibrium of G is a Negotiated Binding Agreement, and so is any profile that Pareto dominates one. Why, then, does the *Coase theorem* not apply, so that bargaining need not reach an efficient outcome?[17] Consider an inefficient Nash equilibrium as a candidate agreement. One might expect that if a player proposed a Pareto-improving action, the other would follow, reading the change as a move to a better outcome for both, and that no inefficient outcome could then survive. But if each correctly believes the other will not move from the Nash proposal, a lone switch to the efficient action is never confirmed, so it changes nothing and does not pay, and agents are free to infer *nothing* from one another's signals. This rests on confirmation being multilateral and, until unanimous, non-binding. A single player's proposal, even one repeated across periods, commits no one and need not be read as a move the others must match.

[17] Note that there are no explicit transfers, so in this respect the environment does not meet the requirements set out by Coase. Nonetheless, even if the underlying game did have transfers the statement may fail in this environment, as discussed in the next subsection.

The inefficiency here comes from the need for mutual confirmation alone, without the frictions that drive it elsewhere. In the baseline there is no cost of delay, information is complete, and agreements are binding, yet the outcome agreed upon can be inefficient. In the bargaining literature, inefficiency instead turns on frictions such as costly delay or an endogenous outside option. Section 7 develops the relation to the bargaining and contracting literatures.

3.3. Discussion of Transferable Utility Games

A natural question, in light of the Coase theorem, is whether allowing transfers can overcome the inefficiencies arising from mutual confirmation. I argue that this is not generally the case, even when players can make transfers contingent on the underlying action profile.

To see this, consider a two-player game with underlying action sets A_i and payoffs $v_i : A \to \mathbb{R}$. Suppose each player i can additionally choose a non-negative contingent transfer schedule $t_i : A \to \mathbb{R}_+$, yielding payoffs $u_i(a, t_i, t_j) = v_i(a) + t_j(a) - t_i(a)$. Since a player can always choose to make no transfers, its best-response payoff against (a_j, t_j) is $\max_{a_i \in A_i}\{v_i(a_i, a_j) + t_j(a_i, a_j)\}$, which is at least $\max_{a_i \in A_i} v_i(a_i, a_j)$. Moreover, this lower bound is attained when the opponent offers no transfers. Thus, introducing transfers does not change the lowest best-response payoffs relevant to the characterisation.

Consequently, while transfers expand the set of feasible payoff allocations, they do not eliminate the constraints arising from mutual confirmation. In particular, the inefficiencies identified in the Cournot duopoly example can persist even when firms can redistribute profits through contingent transfers. The same reasoning concerning best-response payoff bounds extends to games with more than two players.

4 Negotiated Binding Agreement Action Profiles for n-Player Games

This section extends the analysis to n-player games. The conditions of the two-player characterisation remain sufficient, but are no longer necessary. The distinction arises because signalling of agreement restricts individual proposals, not their joint profile: each player's action may be agreeable without the combination being so. I establish sufficient conditions based on individual punishments and necessary conditions based on iterated deletion of individually irrational actions.

4.1. Sufficiency

The sufficient conditions of Theorem 1 extend directly to n-player games.

Theorem 2. *Let G be any underlying game, and suppose there exist $a^*, \underline{a}^1, \ldots, \underline{a}^n \in A$ such that, for $i, j \in N$ with $i \neq j$,*

1. $\underline{a}^i_i \in \arg\max_{a_i \in A_i} u_i(a_i, \underline{a}^i_{-i})$,
2. $u_i(\underline{a}^j) \geq u_i(\underline{a}^i)$,
3. $u_i(a^*) \geq u_i(\underline{a}^i)$.

Then a^* is a Negotiated Binding Agreement action profile.

The proof follows the sufficiency argument of Theorem 1 and is omitted.

4.1.1. No Delay Equilibria

The conditions of Theorem 2 become necessary if attention is restricted to equilibria in which agreement is reached immediately from every history. Following Chatterjee et al. (1993), I call these *No Delay* equilibria.

Definition 1 (No Delay Subgame Perfect Equilibrium)**.** *A strategy profile s^* is a No Delay Subgame Perfect Equilibrium if it is a Subgame Perfect Equilibrium and, following every history $h \in H$, its prescribed profile $s^*(h)$ is confirmed at the first opportunity.*

Proposition 1. *Let A_i be a compact subset of a metric space and u_i continuous for all $i \in N$. If a^* is supported by a No Delay Subgame Perfect Equilibrium, then there exist $\underline{a}^1, \ldots, \underline{a}^n \in A$ satisfying conditions 1–3 of Theorem 2.*

The proof is in Appendix A.

Theorem 2 establishes that these conditions are sufficient for an NBA, while Proposition 1 makes them necessary for No Delay equilibria. They need not characterise No Delay outcomes, since immediate agreement imposes additional off-path restrictions.

Before turning to the necessary conditions for general Negotiated Binding Agreements, I use the following example to illustrate their logic.

4.2. Preview of Necessary Conditions for n-Player Games

Let G be a three-player single-unit first-price auction with heterogeneous valuations. There are three bidders, $N = \{1, 2, 3\}$, where bidder i values the good at v_i, with $v_1 = 6$, $v_2 = 5$, and $v_3 = 2$. Each bidder may bid an integer $b_i \in \{0, 1, \ldots, 7\}$.[18] The highest bidder wins

[18] The maximal bid of 7 plays no role in the analysis. A maximum bid is included only to keep payoffs bounded.

the good, with ties broken uniformly, and pays their bid. Bidders who do not win receive a utility of 0, so utility is the probability of winning times the surplus value. Formally,

$$u_i(b) = \begin{cases} \frac{v_i - b_i}{|\arg\max_{j \in \{1,2,3\}} b_j|} & \text{if } i \in \arg\max_{j \in \{1,2,3\}} b_j \\ 0 & \text{if } i \notin \arg\max_{j \in \{1,2,3\}} b_j \end{cases}$$

Can any bidder agree to the maximal bid, $b_i = 7$, in a Negotiated Binding Agreement? Bidding 7 wins with positive probability at a price above every valuation, giving a strictly negative payoff. A bidder can do better by proposing their own valuation, $s_i(h) = v_i$ for all h, which guarantees a payoff of 0 whether or not the negotiation ends in agreement. Bidding 7 is therefore *individually irrational* in the underlying game, formalised in the next section, so it cannot be part of a Negotiated Binding Agreement, and by signalling of agreement no agent ever proposes it.

Now consider whether bidders 2 or 3 could agree to bid 6. Since no one bids 7, bidding 6 now wins with positive probability, and as both value the good below 6 this gives them a strictly negative payoff. Proposing their own valuation again guarantees 0. So, once 7 is removed, bidding 6 is *individually irrational* for bidders 2 and 3, and by the same argument it is neither part of a Negotiated Binding Agreement nor ever proposed.

The induction continues. Once bidders 2 and 3 do not bid 6, bidder 1 never bids 6 either, since it can then win at a lower price. And bidder 3 never bids 5, which is likewise *iteratively individually irrational*.

By the same argument, any Negotiated Binding Agreement gives bidders 2 and 3 at least 0. Bidder 1 receives at least 1/2. In the worst case for bidder 1, bidders 2 and 3 bid their highest surviving bids, 5 and 4, in every round, and bidder 1 can then bid 5 and tie with bidder 2, guaranteeing a payoff of 1/2.

Since the sufficient conditions deliver the same set, this fully characterises the Negotiated Binding Agreements of this example. The two sets of conditions coincide here, but their logic differs. The sufficient conditions rest on player-specific punishments, the necessary ones on iteratively ruling out outcomes that could never be agreed to.

4.3. Necessary Conditions

I now show that any action proposed at some history must survive a procedure I call *iterated deletion of individually irrational actions*, which to my knowledge is new. The procedure is inductive. An action is individually irrational when, even in its best case, it pays the player less than they can guarantee against the opponents' surviving actions. These actions are deleted, the surviving set shrinks, and the step is repeated until no further action is

removed. The formal definition follows.

Definition 2 (Individually Irrational actions given $C_{-i} \subseteq A_{-i}$)**.** *For a game G, $a_i \in A_i$ is individually irrational given $C_{-i} \subseteq A_{-i}$ if there is a response $\phi_i : C_{-i} \to A_i$ with*

$$\inf_{a_{-i} \in C_{-i}} u_i(\phi_i(a_{-i}), a_{-i}) > \sup_{a_{-i} \in C_{-i}} u_i(a_i, a_{-i}),$$

or equivalently if

$$\inf_{a_{-i} \in C_{-i}} \sup_{a_i' \in A_i} u_i(a_i', a_{-i}) > \sup_{a_{-i} \in C_{-i}} u_i(a_i, a_{-i}).$$

Denote the set of actions that are individually irrational given C_{-i} by $D_i(C_{-i})$.

Definition 3 (Iterated Deletion of Individually Irrational Actions)**.** *For a game G, let $\tilde{A}_i^0 = A_i$ for all $i \in N$. Let $\tilde{A}_{-i}^0 = A_{-i}$. Then for all $m > 0$ let $\tilde{A}_i^m = \tilde{A}_i^{m-1} \backslash D_i(\tilde{A}_{-i}^{m-1})$ where $\tilde{A}_{-i}^{m-1} = \times_{j \neq i} \tilde{A}_j^{m-1}$.*

The set of actions that survive iterated deletion of individually irrational actions, or those that are iteratively individually rational, for i is given by $IIR_i = \bigcap_{m \geq 0} \tilde{A}_i^m$. Let $IIR = \times_{i \in N} IIR_i$.

Two choices in the deletion are worth noting. The maximum (supremum) is over all of A_i, not the surviving actions, because a player may deviate to any action and can secure that payoff through perpetual disagreement, playing it in every period. The action that does so need not survive, as it is a deviation and not an agreement. The opponents, by contrast, are confined to their surviving actions, so no action is deleted using an opponent profile already removed. Restricting the maximum to the surviving actions would change nothing, since, where the supremum is attained, as in the finite and compact, continuous games considered here, the action attaining it is itself individually rational.[19]

The concept of individually irrational actions is related to the *absolute dominance* of Salcedo (2017), developed simultaneously by Halpern and Pass (2018), who compare the worst case of one action with the best case of another, whereas I compare the best case of an action with the inf-sup guarantee.[20] Because the guarantee lets the best response vary with the opponents' play, iterated deletion of individually irrational actions removes at least as much, so its surviving set is at least as small.

[19] The subtlety of a guarantee that is not itself an agreement arises for coalitions, which I return to in Section 6.

[20] Absolute dominance was more recently used by Doval and Ely (2020), who extend it to incomplete information.

Remark 3. *Suppose each $A_i = \Delta(\bar{A}_i)$ is a mixed-strategy space over a finite set of base actions $\bar{A}_i$. For two-player games, the set of individually irrational actions coincides, on pure actions, with those that are absolutely dominated as defined by Salcedo (2017). Specifically, a base action $\bar{a}_i$ is individually irrational if and only if there exists $\sigma_i \in \Delta(\bar{A}_i)$ such that*

$$\min_{\bar{a}_{-i} \in \bar{A}_{-i}} u_i(\sigma_i, \bar{a}_{-i}) > \max_{\bar{a}_{-i} \in \bar{A}_{-i}} u_i(\bar{a}_i, \bar{a}_{-i}).$$

The equivalence follows from the minimax theorem. For more than two players, the corresponding individual-rationality condition considers independent mixtures of opponents' actions, rather than arbitrary correlated distributions.

The first necessary condition is that every equilibrium proposal survives iterated deletion of individually irrational actions in the underlying game, at every history on or off the path. This is the process used to rule out bids in the first-price auction above.

Theorem 3. *If s^* is a Subgame Perfect Equilibrium respecting signalling of agreement, then $s_i^*(h) \in IIR_i$ for all $i \in N$ and $h \in H$.*

The set IIR is non-empty in a large class of games. It contains every action that survives iterated elimination of never best responses to pure actions, a refinement of rationalizability (Bernheim, 1984; Pearce, 1984),[21] and that set is non-empty in any finite game. It may also be strictly larger, since an action that is never a best response can still be individually rational.[22] The prisoners' dilemma illustrates both points.

Example 1. Let the underlying game, G, be the following prisoners' dilemma.

1\2	C	D
C	3,3	0,4
D	4,0	1,1

D is strictly dominant for both players, yet every action profile survives in IIR. The best payoff a player can obtain by playing C is 3, which exceeds the individually rational payoff of 1, so C is not individually irrational. ▼

More generally, every pure Nash equilibrium of the underlying game survives in IIR.

The next result sharpens the necessary conditions, relating Negotiated Binding Agreement payoffs to iterated individual rationality in the underlying game.

[21] As all proposals are pure, the notion is defined with respect to pure actions. When the underlying game admits mixed actions it coincides with *independent* rationalizability in the sense of Bernheim (1984) rather than the correlated version of Pearce (1984): a profile of surviving mixed strategies induces a product belief, with no correlating device available. For two players the two coincide, so the distinction bites only with three or more players, paralleling the independent-mixing qualification to absolute dominance in Remark 3.

[22] A formal statement and proof of the inclusion are given in an earlier working version of the paper.

Theorem 4. *If s^* is a Subgame Perfect Equilibrium respecting signalling of agreement, then:*

$$U_i(s^*|h) \geq \inf_{a'_{-i} \in IIR_{-i}} \sup_{a'_i \in A_i} u_i(a'_i, a'_{-i})$$

for all $h \in H$ and $i \in N$.

Therefore if a^* is a Negotiated Binding Agreement action profile then:

$$u_i(a^*) \geq \inf_{a'_{-i} \in IIR_{-i}} \sup_{a'_i \in A_i} u_i(a'_i, a'_{-i})$$

for all $i \in N$.

I illustrate the use of this result with the same underlying prisoners' dilemma game as in example 1.

Example 1. revisited Consider the underlying game, G, to be that of Example 1.

No actions are individually irrational for any player, as previously argued. The min-max payoff for each player is 1, since the worst case is the opponent playing D. We conclude that (D, C) and (C, D) cannot be a Negotiated Binding Agreement, though the necessary conditions do not rule out (C, C). ▼

Remark 4. *For any underlying game, G,*

$$\inf_{a'_{-i} \in IIR_{-i}} \sup_{a'_i \in A_i} u_i(a'_i, a'_{-i}) \geq \inf_{a_{-i} \in A_{-i}} \sup_{a_i \in A_i} u_i(a_i, a_{-i}).$$

The section echoes the analysis of infinitely repeated games, where agreements must respect individual rationality, but the iterated version here can be far more restrictive. In the first-price auction, individual rationality requires only that each bidder receive a non-negative payoff, whereas iterated deletion also guarantees bidder 1 a surplus of at least 1/2.

The sufficient conditions of Theorem 2 and the necessary conditions of Theorem 4 need not coincide for n-player games, so I do not claim a full characterisation. They do coincide, however, for the class of games identified below, which includes the first-price auction example.

Corollary 1. *If a^{NE} is a pure Nash equilibrium of the underlying game G such that:*

$$u_i(a^{NE}) = \min_{a_{-i} \in IIR_{-i}} \max_{a_i \in A_i} u_i(a_i, a_{-i})$$

i.e. the IIR min-max profiles are mutual, then a^* is a Negotiated Binding Agreement if and only if $u_i(a^*) \geq u_i(a^{NE})$.

This is a direct implication of Theorems 2 and 4. It identifies a class of games in which the Negotiated Binding Agreements are exactly the action profiles that Pareto dominate a Nash equilibrium of the underlying game, namely those games in which a Nash equilibrium delivers each player the individually rational payoff over the set of actions surviving iterated deletion of individually irrational actions. This is the case in the three bidder first price auction used as a leading example for this section.

5 Discounting and Costly Delay

The model so far has treated delay as costless. I now consider whether the results survive when players are impatient. I focus on two-player games, where the original model provides a full characterisation. I establish sufficient conditions for agreement and a necessary lower bound on equilibrium payoffs, before returning to the Cournot example to show that the original punishment payoffs can be approached under discounting.

5.1. The Discounted Game

I normalise the payoff to never reaching agreement to zero, and assume it lies weakly below every payoff of the underlying game, so that $u_i(a) \geq 0$ for all i and a.

Time is divided into periods of length Δ. Players discount at rate $r > 0$, giving a discount factor $\beta = e^{-r\Delta}$ between periods. Each period consists of two rounds of simultaneous proposals. If the profile proposed in the second round repeats that of the first, it becomes binding. Otherwise negotiation continues. An agreement on a reached in period t, counting the initial period as $t = 0$, pays player i the amount $\beta^t u_i(a)$.

Confirmation is imperfect. A player intending to reject the first-round profile instead repeats its proposal by mistake, independently with probability

$$p = 1 - e^{-\lambda\Delta},$$

where $\lambda > 0$ is the tremble rate. For $p > 0$, agreement is reached in finite time almost surely, even if both players always intend to reject.

As before, an agreement is a profile the players intend to confirm, rather than one made binding only by mistake. Good faith therefore requires that equilibrium proposals belong to agreements reached intentionally under the same strategies. Mistakes do not enlarge the set of agreement actions.

Since payoffs are bounded and discounted, the game is continuous at infinity, so subgame perfection can be verified using one-shot deviations at either round of the protocol.

5.2. Sufficiency

The logic of individual punishments continues to apply, but imperfect confirmation introduces an additional incentive constraint. Suppose player j is carrying out the punishment $\underline{a}^i$ and makes an unexpected proposal. In the undiscounted model, player i can reject this proposal without loss: if j rejects too, the proposal cannot bind, regardless of i's response. With imperfect confirmation, however, j may confirm by mistake, giving i an incentive to confirm an attractive proposal.

This can be addressed by introducing an intermediate agreement b^i. Following a deviation by the punisher, play moves to b^i, which the punished player prefers to its own punishment. The punisher must prefer b^i to being punished itself, but prefer its original punishment to b^i, so that initiating the deviation is not profitable.

Proposition 2. *Suppose payoffs are non-negative and there exist $a^*, \underline{a}^1, \underline{a}^2, b^1, b^2 \in A$ such that $u_i(\underline{a}^i) > 0$ for each $i \in \{1,2\}$, each player has at least two distinct actions among these profiles, and, for $i, j \in \{1,2\}$ with $i \neq j$,*

1. $\underline{a}^i_i \in \arg\max_{a_i \in A_i} u_i(a_i, \underline{a}^i_{-i})$,
2. $u_i(a^*) > u_i(\underline{a}^i)$,
3. $u_i(b^i) > u_i(\underline{a}^i)$, *and*
4. $u_j(\underline{a}^j) < u_j(b^i) < u_j(\underline{a}^i)$.

Then there exists $\bar{\beta} < 1$ such that, for every $\beta \in (\bar{\beta}, 1)$, a^* is a Negotiated Binding Agreement of the discounted game whenever the tremble probability p is sufficiently small.

Conditions 1 and 2 retain the best-response and target-payoff requirements of the original characterisation, while condition 4 implies the original willingness-to-punish condition. The remaining conditions ensure that an intermediate agreement provides an incentive to reject a punisher's deviation without rewarding the punisher for making it.

Importantly, the intermediate-agreement conditions are unnecessary with perfect confirmation: when the other player rejects, unilateral confirmation cannot make a proposal binding, so the original construction can rely on weak incentives to reject. For any $p > 0$, those indifferences may instead create profitable deviations. The intermediate agreements provide strict incentives, however small, which suffice when mistakes are rare. Thus the

additional restrictions concern the credibility of off-path rejection under imperfect confirmation, rather than changing the underlying logic of individual punishments.

5.3. Necessity

I now turn to the lower bound on equilibrium payoffs. Whatever the opponent proposes, a player can respond to that proposal and attempt to confirm it. If the opponent intends to reject, it nevertheless confirms by mistake with probability p, so the player can secure agreement after a geometric wait.

For an equilibrium s^*, let $A^*_{-i}(s^*)$ denote the opponent's agreement actions, and define

$$F_i^* := \inf_{a_{-i} \in A^*_{-i}(s^*)} \sup_{a_i \in A_i} u_i(a_i, a_{-i}).$$

Since payoffs are non-negative, $F_i^* \geq 0$. The expected discount factor under a geometric wait is

$$\varphi := \sum_{t=0}^{\infty} (1-p)^t p \beta^t = \frac{p}{1-(1-p)\beta}.$$

Theorem 5. *In any equilibrium of the discounted game respecting signalling of agreement, player i's continuation payoff at the beginning of any period is at least*

$$\varphi F_i^* \geq 0,$$

and is strictly positive whenever $F_i^* > 0$.

The bound need not be sharp, and discounting need not preserve the undiscounted payoff floor. It nevertheless shows that a player cannot be held to the value of never agreeing whenever every agreement action of the opponent leaves it a strictly positive best-response payoff.

5.4. The Cournot Example Under Discounting

I return to the Cournot duopoly to show how the original results are approximated. First, I normalise payoffs by taking

$$u_i(q) = \pi_i(q) + bc_i,$$

so that every payoff is non-negative and never agreeing is worth zero. This normalisation matters under discounting because delayed payoffs are compared with a fixed disagreement value, whereas the undiscounted conditions are unaffected by adding constants to payoffs.

The original punishments do not satisfy the strict conditions above. However, under strict cost asymmetry and strict viability, they can be approximated by punishments that do. The appendix gives the construction, perturbing the original punishments slightly to meet the strict sufficient conditions while their payoffs approach the original floors, 0 for the high-cost firm and $(c_1 - c_2)^2$ for the low-cost firm. Consequently, any Cournot agreement strictly above the original punishment floors can be sustained for sufficiently high β and sufficiently rare mistakes. The Cournot–Nash agreement and a neighbourhood of it are sustained in this way.

Therefore, the original inefficiency is not an artefact of costless delay.

5.5. More Than Two Players

The discounted results above concern two-player games. With several opponents, a player cannot force agreement through the accidental confirmation of a single opponent: all must confirm the same profile. If each intends to reject independently, the probability that all confirm by mistake is p^{n-1}, which becomes small relative to discounting as proposals grow frequent. The two-player payoff floor therefore does not extend directly.

The sufficiency construction likewise uses the ability to respond to one opponent's proposal, and I do not claim its extension here.

6 Coalitional Deviations

Throughout, agreements have been protected only against unilateral deviation. Players who negotiate together, though, may also commit together, holding to a joint plan that none of them would keep alone. I now let the players divide into coalitions, each negotiating and committing as a single block, and ask which agreements survive when deviation, like agreement, can be joint.

I take the coalitions to partition the players: each belongs to exactly one block, and the blocks together make up N. The two familiar cases are the extremes. When every block is a singleton the coalitions are the individual players and the analysis is that of Section 4. When the one block is all of N an agreement must withstand joint deviation by the grand coalition. Between them lie the genuine coalition structures, some players acting together and others apart. The results below hold more generally, for any cover of N by permissible coalitions, but overlap between coalitions raises difficulties I return to in Appendix B. The partition is the natural case, and the one I develop.

A block may deviate to any joint plan its members can agree on. I impose nothing on how they reach it, whether by veto, side payment or prior commitment, since each such

assumption carries its own predictions and I want ones that do not rest on the internal workings of a coalition. The equilibrium I look for is accordingly the most demanding of its kind: an agreement that no permissible block can upset, whatever joint deviation it might contemplate.

6.1. Definitions

A *coalition configuration* $\mathcal{C}$ is the collection of coalitions that may make a binding agreement within the negotiation. I take $\mathcal{C}$ to be a partition of N: every player lies in exactly one $C \in \mathcal{C}$, which I call that player's coalition and term permissible. The constructions below ask only that $\mathcal{C}$ cover N, that every player lie in some permissible coalition, and I develop them for a partition, returning to general covers in Appendix B.

For a non-empty coalition $C \in \mathcal{C}$, let $a_C = (a_i)_{i \in C}$, $A_C = \times_{i \in C} A_i$, $s_C = (s_i)_{i \in C}$ and $S_C = \times_{i \in C} S_i$. Let $a_{-C} = (a_i)_{i \notin C}$, $A_{-C} = \times_{i \notin C} A_i$, $s_{-C} = (s_i)_{i \notin C}$ and $S_{-C} = \times_{i \notin C} S_i$. For a set $B \subset A$, which may or may not have a product structure, let $B_C = \{a_C \in A_C | \exists a'_{-C} \in A_{-C} \text{ s.t. } (a_C, a'_{-C}) \in B\}$ and $B_{-C} = \{a_{-C} \in A_{-C} | \exists a_C \in A_C \text{ s.t. } (a_C, a_{-C}) \in B\}$.

I now extend Subgame Perfect Equilibrium to allow coalitions to deviate jointly. A $\mathcal{C}$-Subgame Perfect Equilibrium requires that at no history can *any* permissible coalition $C \in \mathcal{C}$ deviate jointly and improve every one of its members.[23]

Definition ($\mathcal{C}$-Subgame Perfect Equilibrium)**.** *s^* is a $\mathcal{C}$-Subgame Perfect Equilibrium if, for all partial histories $h \in H$, there does not exist a non-empty coalition $C \in \mathcal{C}$ and a joint strategy $s_C \in \times_{i \in C} S_i$, such that $u_i(s_C, s^*_{-C} | h) > U_i(s^* | h)$ for all $i \in C$.*

This concept sits among familiar ones. For a partition it is the extensive-form counterpart of the coalitional equilibrium of Ray and Vohra (1997), with the blocks of the partition as the deviating units. When every block is a singleton, $\mathcal{C} = \{\{i\} : i \in N\}$, it coincides with the Subgame Perfect Equilibrium of Selten (1965).[24]

To find the set of *$\mathcal{C}$-Negotiated Binding Agreements* I take the $\mathcal{C}$-Subgame Perfect Equilibria of the negotiation game and require a coalitional signalling-of-agreement condition, as before.

Definition 4 ($\mathcal{C}$-Negotiated Binding Agreement)**.** *a^* is a $\mathcal{C}$-Negotiated Binding Agreement action profile if there is a strategy profile s^* of the negotiation game such that*

1. *$a(s^* \mid \emptyset) = a^*$, with $(s^* \mid \emptyset) \in Z'$.*

[23] In essence, this is assuming that, at any history, any permissible coalition may write a private binding agreement that dictates the behaviour they will take going forward. If the agreements were public, the concept would be closer to a coalitional version of Tennenholtz (2004)'s program equilibrium.

[24] When $\mathcal{C}$ is permitted to be a cover and singletons are included this concept generally refines SPE.

2. *s^* is a $\mathcal{C}$-Subgame Perfect Equilibrium.*

3. *s^* respects coalitional signalling of agreements, i.e. for all $C \in \mathcal{C}, h \in H$ $s^*_C(h) \in A^*_C(s^*)$ where*

$$A^*_C(s^*) = \{a_C \in A_C | a_C = a_C(s^*|h) \text{ for some } h \in H \text{ such that } (s^*|h) \in Z'\}$$

is the set of all possible agreement outcomes for coalition $C \in \mathcal{C}$.

What follows contains the conditions of Section 4 as its singleton case, where each player defends the agreement alone. With larger blocks the members are committed to a joint plan, so the test is joint deviation by the block rather than unilateral deviation by its members: this replaces the individual constraints inside a block rather than adding to them, and so need not tighten the conditions on the agreement.

6.2. $\mathcal{C}$-Negotiated Binding Agreement Outcomes

6.2.1. Necessary Conditions

I first show that in any $\mathcal{C}$-Negotiated Binding Agreement the action profile must survive a procedure of *iterated deletion of coalitionally irrational actions* in the underlying game, which generalises the individual deletion of Section 4 by taking a coalition in $\mathcal{C}$, rather than a single player, as the unit of deviation. The procedure provides a recursively justified restriction on the β-core of Aumann (1961), in which the punishments must themselves be justified, and so speaks to the question of Scarf (1971) of a notion of the core for strategic-form games.[25]

Definition 5. *For any underlying game G, a coalition C, and a set $B_{-C} \subseteq A_{-C}$, a joint action $a_C \in A_C$ is coalitionally irrational with respect to B_{-C} if there is a response $\phi_C : B_{-C} \to A_C$ with*

$$\inf_{a_{-C} \in B_{-C}} u_i(\phi_C(a_{-C}), a_{-C}) > \sup_{a_{-C} \in B_{-C}} u_i(a_C, a_{-C}) \qquad \forall i \in C.$$

Denote the set of such joint actions by $D_C(B_{-C})$.

This is the coalitional counterpart of the individual notion in Section 4: for a singleton the response ϕ_i is simply i's best reply and the condition is exactly the individual one, while

[25] Chakrabarti (1988) gives a different answer, taking the punishments to be those that cannot be coalitionally dominated for any action.

for a larger coalition the single response ϕ_C must raise the payoff of every member at once, which no player-by-player benchmark can express.

For a partition the deletion is the procedure of Section 4 run block by block. The blocks are disjoint, so a joint action is deleted only through its own block's comparison, and, where the guarantee is attained, the response attaining it is itself coalitionally rational and survives. Whether that response is drawn from the survivors or from all of A_C therefore makes no difference, exactly as for a single player. So $ICIR(\mathcal{C})$ is the product of the survivors of the blocks, each found as in the individual case.

Definition 6 (Iterated Deletion of Coalitionally Irrational Actions with Respect to $\mathcal{C}$)**.** *For any game G, let $\tilde{A}^0(\mathcal{C}) = A$. For $m > 0$ let:*

$$\tilde{A}^m(\mathcal{C}) = \tilde{A}^{m-1}(\mathcal{C}) \setminus \left[\bigcup_{C \in \mathcal{C}} \left[[D_C(\tilde{A}^{m-1}(\mathcal{C})_{-C})] \times A_{-C} \right] \right]$$

Let the set of action profiles that survive iterated deletion of coalitionally irrational actions, or those that are iteratively coalitionally rational, with respect to $\mathcal{C}$ be denoted by $ICIR(\mathcal{C})$ where $ICIR(\mathcal{C}) = \bigcap_{m>0} \tilde{A}^m(\mathcal{C})$.

Whether $ICIR(\mathcal{C})$ is non-empty is a separate question from the discipline it imposes. Any profile that no permissible coalition can overturn, with its outsiders held at the profile, survives the deletion: a coalition able to beat such a profile's guarantee over the survivors could beat it at the profile's own outsiders, which those outsiders rule out. For a partition this is a coalitional equilibrium in the sense of Ray and Vohra (1997), so $ICIR(\mathcal{C})$ is non-empty wherever one exists, and the existence theory for those equilibria carries over directly. For the richest cover it is a strong Nash equilibrium of G. Neither is necessary, and a sharp characterisation is as delicate as non-emptiness of the β-core the deletion refines, which Scarf (1971) ties to a balancedness condition.

Lemma 2. *If no permissible coalition $C \in \mathcal{C}$ has a joint action a'_C with $u_i(a'_C, a^*_{-C}) > u_i(a^*)$ for all $i \in C$, then $a^* \in ICIR(\mathcal{C})$. In particular every strong Nash equilibrium of G lies in $ICIR(\mathcal{C})$ for every $\mathcal{C}$, and for a partition every coalitional equilibrium of Ray and Vohra (1997) does.*

The deletion that disciplined Negotiated Binding Agreements in Section 4 disciplines their coalitional counterpart in the same way, now through $ICIR(\mathcal{C})$: every proposal a $\mathcal{C}$-Subgame Perfect Equilibrium makes, at any history, must survive it.

Theorem 6. *For any $\mathcal{C}$-Subgame Perfect Equilibrium s^* satisfying coalitional signalling of agreement, $s^*(h) \in ICIR(\mathcal{C})$ at every history h.*

When the grand coalition is permissible this restriction is already strong: nothing coalitionally irrational for N can be proposed, so only weakly Pareto optimal profiles remain.

The floor of Theorem 4 has a coalitional analogue: at every history a $\mathcal{C}$-Negotiated Binding Agreement must leave no coalition able to guarantee all its members more than the agreement gives them. I state it through the β-core of Aumann (1961), taken with respect to $ICIR(\mathcal{C})$.

Definition 7. *$a^* \in A$ is in the β-core with respect to $ICIR(\mathcal{C})$ if there is no $C \in \mathcal{C}$ and response $\phi_C : [ICIR(\mathcal{C})]_{-C} \to A_C$ such that $\inf_{a_{-C} \in [ICIR(\mathcal{C})]_{-C}} u_i(\phi_C(a_{-C}), a_{-C}) > u_i(a^*)$ for all $i \in C$.*

In the standard β-core, a profile is undominated if no coalition can guarantee all its members more by deviating, with the others taken to hold to some fixed profile drawn from all of A_{-C}. The punishments that enforce it need not themselves be justifiable. The β-core with respect to $ICIR(\mathcal{C})$ confines those punishments to $[ICIR(\mathcal{C})]_{-C}$, the profiles that survive deletion, so the threats deterring a deviation are consistent in the same recursive sense as the solution they support.

Theorem 7. *In any $\mathcal{C}$-Subgame Perfect Equilibrium s^* satisfying coalitional signalling of agreement, for any history h and any coalition $C \in \mathcal{C}$ there is no response $\phi_C : [ICIR(\mathcal{C})]_{-C} \to A_C$ such that*

$$\inf_{a_{-C} \in [ICIR(\mathcal{C})]_{-C}} u_i(\phi_C(a_{-C}), a_{-C}) > U_i(s^*|h) \qquad \text{for all } i \in C.$$

In other words, whenever the continuation from h reaches agreement, the agreed profile $a(s^*|h)$ lies in the β-core with respect to $ICIR(\mathcal{C})$.

An outcome can be both Pareto efficient and individually rational in the underlying game and still fail to be a $\mathcal{C}$-Negotiated Binding Agreement when $\{N, \{i\}_{i \in N}\} \subseteq \mathcal{C}$.

I now turn to sufficient conditions.

6.2.2. Sufficient Conditions

The sufficient conditions rest on the underlying game G and, like those of Theorem 2, turn on the identity of the deviating coalition rather than on the deviation it makes. The two-player construction held an agreement in place with a punishment in which the punished player best responded, so that, the others fixed, it had nothing to gain by moving. The coalitional counterpart asks as much of a single member of each coalition, and that member may differ across the profiles the punishment holds in place.

Theorem 8. *Suppose $\mathcal{C}$ is a partition of N. Take $a^* \in A$ and, for each $C \in \mathcal{C}$, a punishment profile $\underline{a}^C \in A$, with $\underline{a}^N = a^*$ if $N \in \mathcal{C}$. For $i \in C$, let*

$$m_i^C = \max_{b_C \in A_C} u_i(b_C, \underline{a}^C_{-C}),$$

the most member i can obtain by a joint deviation of C while the other coalitions hold their part of $\underline{a}^C$. Suppose that, for every $C \in \mathcal{C}$ and every $a \in \{a^*\} \cup \{\underline{a}^D : D \in \mathcal{C}\}$, there is some $i \in C$ with

$$u_i(a) \geq m_i^C.$$

Then a^* is a $\mathcal{C}$-Negotiated Binding Agreement.

The member meeting the bound may differ across these profiles. At $\underline{a}^C$ it can hold only with equality, since $u_i(\underline{a}^C) \leq m_i^C$ always, so there it requires a member who best responds in C's own punishment. At a^* and at the other punishments it instead asks that some member not gain by triggering C's punishment, and this need not be the same member.

A second, simpler condition drops the punishments entirely, at the cost of a stance on breakdowns. Call a^* a *coalitional equilibrium* of G with respect to $\mathcal{C}$ if no permissible coalition C has a joint action a_C with $u_i(a_C, a^*_{-C}) > u_i(a^*)$ for all $i \in C$: holding the others at a^*, no coalition can raise all of its members.

Remark 5. *If perpetual disagreement is valued at the* lim inf *of the proposed payoffs, every coalitional equilibrium of G with respect to $\mathcal{C}$ is a $\mathcal{C}$-Negotiated Binding Agreement. The proof is given in Appendix A.*

Every strong Nash equilibrium of G is a coalitional equilibrium for every $\mathcal{C}$, and for a partition so is every coalitional equilibrium in the sense of Ray and Vohra (1997). Remark 5 sustains them all.

Both results connect the sustainable agreements to the β-core of Aumann (1961). There a coalition is deterred by holding the others to a profile that need not itself be justifiable. Here the profiles used as punishments are themselves agreements the sustaining strategies reach, and under Remark 5 the agreement itself is the only prescribed profile. The threats are therefore drawn from proposals that the same negotiating strategies could bring to agreement.

6.3. A Buyers' Group

Consider two buyers, 1 and 2, who each need one unit of an input, worth v and costless to supply. Each buyer chooses whether to source from a common supplier, player 3, or an

outside source, $A_1 = A_2 = \{S, O\}$. The supplier chooses a price regime, $A_3 = \{H, L\}$, with $p_L < p_H < v$. The outside source is viable only if both buyers use it, giving each a payoff w. A buyer turning to it alone receives nothing. Writing j for the other buyer, payoffs are

$$u_i(a) = \begin{cases} v - p_{a_3}, & a_i = S, \\ w\,\mathbf{1}[a_j = O], & a_i = O, \end{cases} \quad (i = 1, 2), \qquad u_3(a) = p_{a_3}\big|\{i : a_i = S\}\big|.$$

Assume $v - p_H < w < v - p_L$, so that the outside source is preferable to the high price but not the low one. For instance, take $v = 10$, $p_H = 9$, $p_L = 6$ and $w = 3$.

The game has two pure Nash equilibria: (S, S, H), where the supplier extracts surplus from both buyers, and (O, O, H), where both buyers use the outside source. Trade at the low price, (S, S, L), is not a Nash equilibrium, since the supplier would raise its price. Under individual negotiation, the extractive outcome (S, S, H) remains a Negotiated Binding Agreement. Each buyer's floor is $v - p_H$, since turning to the outside source alone yields nothing. Although the boycott (O, O, H) can also be agreed upon, individual negotiation does not rule out extraction.

Now suppose the buyers negotiate as a coalition, $\mathcal{C} = \{\{1, 2\}, \{3\}\}$. Together they can guarantee each member w by turning to the outside source, regardless of the supplier's action. This option survives $ICIR(\mathcal{C})$, since against the high price no joint deviation improves both buyers' payoffs. By Theorem 7, no $\mathcal{C}$-Negotiated Binding Agreement can give both buyers strictly less than w. Since both receive $v - p_H < w$ at (S, S, H), the extractive agreement is ruled out.

Trade at the low price, (S, S, L), satisfies this restriction and is sustainable by Theorem 8. If the supplier raises its price, the buyers can jointly turn to the outside source, a credible punishment that leaves each with $w > v - p_H$. The boycott (O, O, H) also remains sustainable, so coalitional negotiation does not guarantee efficiency. It does, however, eliminate the extractive agreement and allow the buyers either to trade on favourable terms or to walk away together.

The source of the coalition's power is that the outside option benefits both buyers simultaneously, even though neither can use it alone. No transfers between buyers are needed, and the threat need never be carried out. This distinguishes a buyers' group from, for instance, a bidding ring competing for an indivisible prize: without transfers, winning the prize cannot improve every member's payoff. Here, joint negotiation creates a common fallback that individual buyers lack, providing countervailing power against the supplier.

7 Literature Review

A number of papers have approached the question of which binding agreements can be made for strategic-form games using an approach close to or inspired by the farsighted stable set of Harsanyi (1974). I instead take a more non-cooperative game-theoretic approach, exploring the SPE in a fully specified negotiation game where agents signal the agreement they would like to take. Within this strand of literature, Mariotti (1997) has the closest model and also considers an explicit negotiation protocol. The extensive form of the negotiation protocol is similar, but the payoff of perpetual disagreement is set to $-\infty$. Mariotti (1997) takes an approach close to the strong Subgame Perfect Equilibrium of Rubinstein (1980). Mariotti also imposes a refinement on this subgame-perfect concept, based on the farsighted stable set. Mariotti (1997) does not provide general conditions for his solution concept, due to the complexity that the history-dependent negotiation entails. Mariotti instead proposes a history-independent version of the solution concept, in line with Harsanyi (1974), where agents' strategies map only from the current proposal to the next, rather than considering all previous proposals. In this history independent version, Mariotti (1997) provides some necessary conditions for agreement outcomes similar to those provided in this paper for both Negotiated Binding Agreements and $\mathcal{C}$-Negotiated Binding Agreements. The paper also provides sufficient conditions for agreement outcomes in a class of two-player games with conditions on the Pareto frontier, similarly using a notion of individual punishments.

Chwe (1994); Xue (1998); Ray and Vohra (2015, 2019) also consider versions of the farsighted stable set. The closest with respect to my paper is Ray and Vohra (2019), which considers games with transferable utility and defines the maximal farsighted stable set, which additionally requires a subgame perfect-like condition, imposing optimality given others' strategies at all histories of the negotiation. They provide general conditions linking the farsighted stable set as defined in Ray and Vohra (2015) to this concept. I instead consider general games, rather than a game with transferable utility, and link the concept of $\mathcal{C}$-Negotiated Binding Agreements to an alternative cooperative game-theoretic notion, the β-core of Aumann (1959, 1961). Finding the farsighted stable set is challenging and some papers have looked at finding the farsighted stable set for a specific underlying game (Suzuki and Muto, 2005; Nakanishi, 2009).

Other papers have also proposed fully non-cooperative models of negotiation over binding agreements for strategic-form games, based on a dynamic game of negotiation. Kalai (1981) studies a fully specified model of negotiation, proposing a non-cooperative extensive-form game. In that model, agents propose an individual action in the underlying game. If an agent changes their proposal within a period then they are no longer permitted to change their proposal again. The process ends at time t with the proposal profile proposed

in that period. Kalai examines the perfect equilibria of Selten (1988) and shows that only cooperation can be sustained in the two-player prisoners' dilemma. Nishihara (2022) has extended this to an n-player prisoners' dilemma, maintaining Kalai's negotiation protocol. Kalai's protocol is close in spirit to the one here, with agents negotiating over the agreement by proposing their own action. Bhaskar (1989) examines a model of pre-play agreement over a symmetric two-player Bertrand game. In a similar sense to this model, agents make proposals of the prices they will take, and have the opportunity to revise their proposals sequentially. Confirmation requires one agent not to change their proposal after seeing the others. Bhaskar (1989) examines the perfect equilibria of such an agreement game and concludes that only the monopoly price can be sustained.

Most directly comparable in its confirmation mechanism is Attanasi et al. (2015), who study bargaining over the strategy profile of a non-cooperative game: players propose a profile, reproposing it counts as confirmation, and a profile left unconfirmed carries to the next round, much as consecutive repetition does here.[26] They restrict attention to two players and obtain a unique and efficient agreement by imposing a weak-dominance argument at the point of confirmation, under which the active player confirms whenever doing so is no worse than its best continuation without confirmation and so selects the branch leading to the Pareto-superior agreement. Their disagreement point is the Nash equilibrium of the underlying game, and subgame perfect equilibria may fail to exist or to be played. In contrast, I characterise sustainable agreements under subgame perfection and the good-faith restriction without imposing that selection criterion, allowing inefficient agreements to persist.

Closely related is the "pledge-and-review" bargaining protocol of Harstad (2022) for public goods games, similar to the one in this paper. In that model, Harstad (2022) shows that when agents confirm by default and discounting is hyperbolic, a folk theorem remains for the subgame-perfect equilibrium outcomes of the game. When considering stationary subgame perfect equilibria, each agent's pledge must be the result of maximising *some* weighted Nash product. However, the weights used by each agent may differ, and therefore many inefficient equilibria arise despite this. In my work, I instead consider a more general class of games and an alternative refinement of SPE. By the construction of the equilibria, imposing stationarity does not substantially change sufficiency.

More broadly, the inefficiency here differs in source from that in the bargaining literature. The alternating-offers model of Rubinstein (1982) yields efficient agreement under constant discounting, but under hyperbolic discounting, or with an endogenous outside option as in Busch and Wen (1995), a folk theorem can be sustained. The latter differs from the

[26] The same confirmation mechanism is taken to the laboratory by the same authors, in a prisoners' dilemma (Attanasi et al., 2013) and in a Hawk–Dove environmental game (Attanasi et al., 2012).

mechanism here, since that outside option need not be consistent with any agreement. With more than two players, regaining efficiency requires further restrictions on the equilibrium (Chatterjee et al., 1993). These inefficiencies turn on delay or on the outside option, whereas in the baseline here delay is costless and the inefficiency lies in the agreement reached.

Other papers take a more cooperative game-theoretic approach to the agreements that can be made for games, for instance strong Nash equilibrium (Aumann, 1959) and the β-core (Aumann, 1959, 1961). In my paper, under the lim inf valuation of disagreement, $\mathcal{C}$-Negotiated Binding Agreement outcomes lie between the β-core and strong Nash equilibrium, and are backed by an explicit negotiation procedure. Given this, my paper can also be seen in the light of the Nash program pointed to in Nash (1953), as the necessary and sufficient conditions $\mathcal{C}$-Negotiated Binding Agreement outcomes can be seen as a perturbed version of the β-core.

Similar concepts also exist, such as the γ-core (Chander and Tulkens, 1997). Chander (2007) provides further justification for the γ-core by showing it is *an* equilibrium to an infinitely repeated game where agents decide whether to cooperate or not in each round. Chander and Wooders (2020) define a notion of coalitional Subgame Perfect Equilibrium for underlying games with transferable utility, where a coalition's deviation payoff is with respect to the best Subgame Perfect Equilibrium assuming all other players act without cooperation. A number of papers have also proposed notions of rationalizability for coalitions in a cooperative sense (Herings et al., 2004; Ambrus, 2006, 2009; Grandjean et al., 2017). The iterated deletion of coalitionally irrational actions here can be seen in this light, though it is distinct from each. A strand of literature abstracts from the negotiation process *within* a group and takes a cooperative perspective, focusing on Pareto undominated actions that prevent new groups from breaking and forming (Ray and Vohra, 1997; Diamantoudi and Xue, 2007).

Several papers consider a form of communication for equilibrium selection (Bernheim et al., 1987; Farrell and Maskin, 1989; Bernheim and Ray, 1989; Rabin, 1994). My paper is related in the sense that agents can communicate via the negotiation procedure, proposing the agreement action they would like, to select the outcome of the underlying game that will be played. However, the perspective is different, as these concepts are about refining a given set of non-binding agreements represented by the (potential mix over) SPE or Nash Equilibria of an underlying game, whereas I allow agents to make a binding agreement of potentially any outcome.

In the contracting literature, the closest work is Jackson and Wilkie (2005); Yamada (2003); Ellingsen and Paltseva (2016), who propose models in which agents all have a strategic input on the *structure* of the contract over an underlying strategic environment.

In a similar way, Negotiated Binding Agreements allows for all agents to have a strategic input on the action they will agree to in the underlying game. On the other hand, Kalai et al. (2010), Peters and Szentes (2012) and Tennenholtz (2004) all consider the possibility of all agents proposing contracts surrounding their own play in an underlying game, where these contracts can be a function of the contracts of others. This allows agents to specify reactions to deviations in full, and can allow for these to be fully specified at a higher level also. In contrast to these, my paper requires that agents only propose actions they could agree to, whereas these papers allow contracts to specify actions that would never be the result of equilibrium. Inefficiency persists, but with a novel lower bound.

The way payoffs are defined for perpetual disagreement can be seen as similar to the literature of infinitely repeated games with no discounting (Aumann and Shapley, 1994; Rubinstein, 1994). The individual punishments in the paper are also similar to the player-specific punishments used in infinitely repeated games (Fudenberg and Maskin, 1986; Abreu et al., 1994). The sufficient conditions I use are more restrictive: player-specific punishment only requires that a punishment give the punished player an individually rational payoff and that each prefer to punish rather than be punished, whereas I also require that the punished player best responds to its punishment in the underlying game. This is because there are no continuation rewards for carrying out a punishment, as there would be in a repeated game, so an agent facing the constant punishment of others must be best responding, since otherwise it could improve on it.

8 Conclusion

I have studied how agents negotiate binding agreements over their actions, where proposals must be consistent with agreements that could be reached and every party must confirm the outcome. Even with complete information and costless delay, inefficient agreements can persist. A player cannot implement a Pareto improvement unilaterally, and the others need not respond to its proposal.

For two players, I provide a full characterisation of sustainable agreements through individual punishments, which must themselves be agreements. These conditions remain sufficient for n players, while iterated deletion of individually irrational actions provides necessary conditions. The inefficiency persists under discounting and imperfect confirmation. Allowing coalitions changes which punishments are credible and can eliminate extractive agreements, although it does not generally restore efficiency.

The central insight is that mutual consent does more than determine how agreements become binding: it also shapes which outcomes can be sustained. Even when agents negotiate in good faith, with no informational frictions or costs of delay, the need for mutual

confirmation can prevent mutually beneficial changes from taking place.

Good faith proves compatible with surprisingly extreme bargaining. Looking at this through the lens of Cournot competition, because what may be proposed is settled within the equilibrium, a firm can demand that its rival be flooded from the market for nothing and still negotiate in good faith, as long as that outcome could itself be agreed. The restriction bites on the threats that would enforce an agreement, not on the ambition of the demands that lead to one.

A Proofs

Proofs for Section 2

Proof of Remark 1: Let $\underline{u}_i \leq u_i(a)$ for all $a \in A$ be player i's disagreement payoff, and fix any a^* with $u_i(a^*) > \underline{u}_i$ for every i. Consider the strategy profile under which all players propose a^* on the path and confirm it, and under which, following any deviation, they propose two fixed distinct profiles in alternation, which at least two actions each make possible, so that no profile is repeated and perpetual disagreement results, paying the deviator $\underline{u}_i$. On the path a^* is confirmed, paying each i the amount $u_i(a^*) > \underline{u}_i$. Any deviation yields $\underline{u}_i$, and since agreement requires unanimous confirmation no player can secure more by departing from either phase. The profile is therefore a subgame perfect equilibrium. As a^* was arbitrary, the claim follows. ■

Proof of lemma 1: Fix $\epsilon > 0$ and write $L = \liminf_{k\to\infty} u_i(a^k)$. By definition of the lim inf there is T with $u_i(a^t) \geq L - \epsilon$ for all $t > T$. For every $\delta \in (0,1)$,

$$(1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}u_i(a^t) = (1-\delta)\sum_{t=1}^{T}\delta^{t-1}u_i(a^t)+(1-\delta)\sum_{t>T}\delta^{t-1}u_i(a^t) \;\geq\; (1-\delta)\sum_{t=1}^{T}\delta^{t-1}u_i(a^t)+(L-\epsilon)\,\delta^T,$$

using $(1-\delta)\sum_{t>T}\delta^{t-1} = \delta^T$. As $\delta \to 1$ the finite sum is multiplied by $(1-\delta) \to 0$ and $\delta^T \to 1$, so the right-hand side tends to $L-\epsilon$. Hence $\liminf_{\delta\to1}(1-\delta)\sum_{t\geq1}\delta^{t-1}u_i(a^t) \geq L-\epsilon$. As ϵ was arbitrary this lim inf is at least $L = \liminf_k u_i(a^k)$. The symmetric argument gives $\limsup_{\delta\to1}(1-\delta)\sum_{t\geq1}\delta^{t-1}u_i(a^t) \leq \limsup_k u_i(a^k)$. Hence any sequence $\delta_m \to 1$ along which $V_i(\delta_m)$ converges has its limit between $\liminf_{\delta\to1} V_i(\delta)$ and $\limsup_{\delta\to1} V_i(\delta)$, and so in $[\liminf_k u_i(a^k), \limsup_k u_i(a^k)]$. ■

Proofs for Sections 3–4

Proof of Theorem 1:

Sufficiency: Write $\underline{a}^j$ for the profile used to punish j, and a^k for the profile proposed

in period k of a history. The strategy s^* carries a standing proposal p, read off the recent history, and meets any departure with one rejection round before a punishment. Starting from $p = a^*$, after a period that realised a^k with standing proposal p:

1. if $a^k = p$, the standing proposal stays p, and if $a^k = a^{k-1} = p$ the profile p is agreed.

2. otherwise let $D = \{i : a_i^k \neq p_i\}$ be the players who departed, and select one $i \in D$, by a fixed tie-break when $|D| = 2$. The next standing proposal is the rejection profile $(p_i, \underline{a}^i_{-i})$, in which i returns to p_i and the other proposes its component of $\underline{a}^i$. If that profile is realised with no further departure, the standing proposal becomes $\underline{a}^i$ from the next period on.

The rule applies again within a rejection or punishment round: a departure sends play through $(p_i, \underline{a}^i_{-i})$ for the selected departer i and on to $\underline{a}^i$, whatever the standing p. Every proposed action is a component of a^*, $\underline{a}^1$ or $\underline{a}^2$, each of which is agreed after some history, so signalling of agreement holds. From any history the continuation reaches agreement after finitely many further proposals unless a player keeps departing, so it suffices to check that no departure pays.

Take any departure from the standing proposal p, and let i be the selected departer, so $a_i^k \neq p_i$. From the next period the other player proposes $\underline{a}^i_{-i}$, and holds it as long as i alone departs. Player i cannot confirm the profile it departed to, (p_{-i}, x_i): a profile binds only when both players repeat it, and the other has moved to $\underline{a}^i_{-i}$. The only profile i can still bind by repeating is $(\underline{a}^i_{-i}, x_i)$, and only when $p_{-i} = \underline{a}^i_{-i}$, where condition 1 bounds it by $u_i(\underline{a}^i_{-i}, x_i) \leq \max_{a_i} u_i(a_i, \underline{a}^i_{-i}) = u_i(\underline{a}^i)$. Otherwise play settles at $\underline{a}^i$. And if i never lets a profile repeat, every later period leaves the other at $\underline{a}^i_{-i}$ and so pays i at most $\max_{a_i} u_i(a_i, \underline{a}^i_{-i}) = u_i(\underline{a}^i)$, whence the disagreement value, between the lim inf and lim sup of these payoffs, is at most $u_i(\underline{a}^i)$ as well. So a departure secures i at most $u_i(\underline{a}^i)$, for any value the interval assigns to perpetual disagreement. Following the prescribed strategy from any history leads to a^* or a punishment $\underline{a}^k$, each of which gives player i at least $u_i(\underline{a}^i)$ by conditions 2 and 3. A unilateral departure therefore makes a player the selected departer and secures it at most its own punishment payoff, no more than it already had. And since the rule applies afresh at every history, including any reached after a simultaneous departure, no deviation pays. So s^* is a subgame-perfect equilibrium respecting signalling of agreement.

Necessity: To see that only such a^* can be sustained, take any a^* that is a Negotiated Binding Agreement outcome in the two-player case given by the SPE s^*. Denote $\tilde{A} = \{a \in A | \exists h \in H \text{ s.t. } s^*(h) = a\}$. Note by signalling of agreement these are the only actions that can be proposed in equilibrium. Therefore $s^*_{-i}(h) \in \tilde{A}_{-i}$ for all $h \in H$. As

s^* is an SPE it must be that there is no profitable deviation. Notice that $U_i(s^*|h) \geq \inf_{a_{-i}\in\tilde{A}_{-i}} \max_{a_i\in A_i} u_i(a_i, a_{-i})$. Suppose not $U_i(s^*|h) < \inf_{a_{-i}\in\tilde{A}_{-i}} \max_{a_i\in A_i} u_i(a_i, a_{-i})$. It follows that $\inf_{a_{-i}\in\tilde{A}_{-i}} \max_{a_i\in A_i} u_i(a_i, a_{-i}) - U_i(s^*|h) > 0$. Consider a deviation to s_i' such that $s_i'(h') = s_i^*(h')$ for all h' such that $h = (h', h'')$ while $s_i'(h')$ is such that $u_i((s_i', s_{-i}^*)(h')) = \max_{a_i\in A_i} u_i(a_i, s_{-i}^*(h'))$ for all other histories. Suppose such a deviation leads to perpetual disagreement. Denote the sequence induced by such a strategy by $z' = (a^1, a^2, ..., a^t, ...)$. Each a_{-i}^t is an agreement action of the other player, drawn from $\tilde{A}_{-i}$, so player i best responds to an element of $\tilde{A}_{-i}$: $u_i(a^t) = \max_{a_i\in A_i} u_i(a_i, a_{-i}^t) \geq \inf_{a_{-i}\in\tilde{A}_{-i}} \max_{a_i\in A_i} u_i(a_i, a_{-i})$, the infimum taken over the agreement actions arising at any history.

By definition:

$$\begin{aligned} U_i(s_i, s_{-i}^*|h) &\geq \liminf_{t\to\infty} u_i(a^t) \\ &\geq \inf_{a_{-i}\in\tilde{A}_{-i}} \max_{a_i\in A_i} u_i(a_i, a_{-i}) \Rightarrow U_i(s_i, s_{-i}^*|h) > U_i(s^*|h) \end{aligned}$$

therefore it cannot be that s^* is an SPE if the deviation ends in perpetual disagreement. The argument for agreement is direct from the definition.

Therefore $U_i(s^*|h) \geq \inf_{a_{-i}\in\tilde{A}_{-i}} \max_{a_i\in A_i} u_i(a_i, a_{-i}) =: g_i$ at every history, where $\tilde{A}_{-i}$ is the set of agreement actions of j. In particular every agreement reached under s^* pays i at least g_i, so $u_i(a^*) \geq g_i$. Since A_{-i} is compact and u_i continuous, the infimum is attained at some $\bar{a}_{-i}$ in the closure of $\tilde{A}_{-i}$. Taking agreements in which j's action approaches $\bar{a}_{-i}$, their profiles have a limit point $\underline{a}^i \in A$ with j playing $\bar{a}_{-i}$. At $\underline{a}^i$ player i receives at least g_i, by the bound and continuity, and at most $\max_{a_i} u_i(a_i, \bar{a}_{-i}) = g_i$, since $\bar{a}_{-i}$ attains the infimum. Hence $u_i(\underline{a}^i) = g_i$ with $\underline{a}_i^i \in \arg\max_{a_i} u_i(a_i, \bar{a}_{-i})$, which is condition 1, and $u_i(a^*) \geq g_i = u_i(\underline{a}^i)$, which is condition 3. Finally $\underline{a}^j$ is itself a limit of agreements, so it too pays i at least $g_i = u_i(\underline{a}^i)$, giving condition 2, $u_i(\underline{a}^j) \geq u_i(\underline{a}^i)$. The required punishments therefore exist. ■

Proof of theorem 3: Suppose, towards a contradiction, that for some history $h' \in H$, $s_i(h') = a_i \in D_i(A_{-i})$. By consistency of agreement signalling, there exists some $h \in H$ such that $a_i(s|h) = a_i$. Therefore,

$$U_i(s|h) = u_i(a(s|h)) \leq \sup_{a'_{-i}\in A_{-i}} u_i(a_i, a'_{-i}).$$

Take

$$\epsilon = \inf_{a'_{-i}\in A_{-i}} \sup_{a'_i\in A_i} u_i(a'_i, a'_{-i}) - u_i(a(s|h)) > 0.$$

Take a function $\phi_i : A_{-i} \to A_i$ such that

$$u_i(\phi_i(a_{-i}), a_{-i}) \geq \sup_{a'_i \in A_i} u_i(a'_i, a_{-i}) - \frac{\epsilon}{2}$$

for all $a_{-i} \in A_{-i}$. Consider a deviation $s'_i(h'') = \phi_i(s_{-i}(h''))$ for all $h'' \in H$. It follows that

$$\begin{aligned} U_i(s'_i, s_{-i}|h) &\geq \inf_{a_{-i} \in A_{-i}} u_i(\phi_i(a_{-i}), a_{-i}) \\ &\geq \inf_{a_{-i} \in A_{-i}} \sup_{a'_i \in A_i} u_i(a'_i, a_{-i}) - \frac{\epsilon}{2} \\ &= U_i(s|h) + \frac{\epsilon}{2} > U_i(s|h). \end{aligned}$$

Thus a profitable deviation exists, contradicting Subgame Perfect Equilibrium. We conclude that $s_i(h) \notin D_i(A_{-i})$ for every $i \in N$ and $h \in H$.

Now suppose inductively that $s_j(h') \in \tilde{A}^m_j$ for all $j \in N$ and $h' \in H$. Suppose, towards a contradiction, that for some $i \in N$ and $h' \in H$, $s_i(h') = a_i \in D_i(\tilde{A}^m_{-i})$. By consistency of agreement signalling, there exists some $h \in H$ such that $a_i(s|h) = a_i$. Since $s_{-i}(h'') \in \tilde{A}^m_{-i}$ for every $h'' \in H$, it follows that

$$U_i(s|h) = u_i(a(s|h)) \leq \sup_{a'_{-i} \in \tilde{A}^m_{-i}} u_i(a_i, a'_{-i}).$$

Take

$$\epsilon = \inf_{a'_{-i} \in \tilde{A}^m_{-i}} \sup_{a'_i \in A_i} u_i(a'_i, a'_{-i}) - u_i(a(s|h)) > 0.$$

Take a function $\phi_i : \tilde{A}^m_{-i} \to A_i$ such that

$$u_i(\phi_i(a_{-i}), a_{-i}) \geq \sup_{a'_i \in A_i} u_i(a'_i, a_{-i}) - \frac{\epsilon}{2}$$

for all $a_{-i} \in \tilde{A}^m_{-i}$. Consider a deviation $s'_i(h'') = \phi_i(s_{-i}(h''))$ for all $h'' \in H$. It follows that

$$\begin{aligned} U_i(s'_i, s_{-i}|h) &\geq \inf_{a_{-i} \in \tilde{A}^m_{-i}} u_i(\phi_i(a_{-i}), a_{-i}) \\ &\geq \inf_{a_{-i} \in \tilde{A}^m_{-i}} \sup_{a'_i \in A_i} u_i(a'_i, a_{-i}) - \frac{\epsilon}{2} \\ &= U_i(s|h) + \frac{\epsilon}{2} > U_i(s|h). \end{aligned}$$

This contradicts Subgame Perfect Equilibrium. Therefore $s_i(h) \notin D_i(\tilde{A}^m_{-i})$ for every $i \in N$ and $h \in H$, and hence $s_i(h) \in \tilde{A}^{m+1}_i$. By induction, $s_i(h) \in IIR_i$ for every $i \in N$ and

$h \in H$. ■

Proof of theorem 4: Suppose not. Then there is some $i \in N$ and $h \in H$ such that

$$\inf_{a'_{-i} \in IIR_{-i}} \sup_{a'_i \in A_i} u_i(a'_i, a'_{-i}) > U_i(s^*|h).$$

It must be that a) s^* is a Subgame Perfect Equilibrium of the negotiation game and b) by Theorem 3, $s^*_{-i}(h') \in IIR_{-i}$ for all $h' \in H$.

Let

$$\epsilon = \inf_{a'_{-i} \in IIR_{-i}} \sup_{a'_i \in A_i} u_i(a'_i, a'_{-i}) - U_i(s^*|h) > 0.$$

Construct $\phi_i : IIR_{-i} \to A_i$ such that

$$u_i(\phi_i(a_{-i}), a_{-i}) \geq \sup_{a_i \in A_i} u_i(a_i, a_{-i}) - \frac{\epsilon}{2}$$

for all $a_{-i} \in IIR_{-i}$.

Consider a deviation to $s'_i(h') = \phi_i(s^*_{-i}(h'))$ for all $h' \in H$. It follows that

$$\begin{aligned} U_i(s'_i, s^*_{-i}|h) &\geq \inf_{a_{-i} \in IIR_{-i}} u_i(\phi_i(a_{-i}), a_{-i}) \\ &\geq \inf_{a_{-i} \in IIR_{-i}} \sup_{a_i \in A_i} u_i(a_i, a_{-i}) - \frac{\epsilon}{2} \\ &= \frac{\inf_{a_{-i} \in IIR_{-i}} \sup_{a_i \in A_i} u_i(a_i, a_{-i}) + U_i(s^*|h)}{2} \\ &> U_i(s^*|h). \end{aligned}$$

A contradiction, as therefore s^* is not a Subgame Perfect Equilibrium of the negotiation game. ■

Proofs for Section 5

Throughout write

$$\mu := \sup_{i \in N} \sup_{a \in A} |u_i(a)| < \infty, \qquad \beta = e^{-r\Delta}, \qquad p = 1 - e^{-\lambda\Delta}, \qquad \varphi = \frac{p}{1 - (1-p)\beta}.$$

One-shot deviations. Since payoffs are bounded by μ and received only at agreement, the continuation payoff beyond period t is bounded by $\beta^t \mu \to 0$. The game is therefore continuous at infinity, and the one-shot-deviation principle applies at both rounds, including with stochastic trembles. Deviations are unrestricted by good faith.

Proof of Proposition 1. Under No Delay every prescribed profile is itself an agreement, so the opponents' jointly prescribed actions occur in agreements and the necessity argument of Theorem 1 applies. Compactness of the action sets and continuity of the payoffs then yield the required punishment profiles. ■

Proof of Proposition 2. Write $a^i = \underline{a}^i$. Conditions 3 and 4 imply, for $i \neq j$,

$$u_i(a^j) > u_i(b^j) > u_i(a^i).$$

Choose $\bar{\beta} < 1$ sufficiently close to one that

$$u_i(a^i) < \beta u_i(a^j)$$

for both players whenever $\beta > \bar{\beta}$.

I construct a strategy using five targets,

$$T = \{a^*, a^1, a^2, b^1, b^2\}.$$

In the first round both players propose the current target $t \in T$. Let x be the realised first-round profile.

If $x = t$, both players confirm. If exactly player i rejects, the next target is a^i. If both reject, the next target is a^*.

If $x \neq t$, both players reject. Each rejection uses a component of a target in T different from the player's current proposal, which is possible by assumption. When both reject, the next target is

$$w = \begin{cases} t, & t \in \{a^*, b^1, b^2\}, \\ a^i, & t = a^i,\ x_i \neq a^i_i,\ x_j = a^i_j, \\ b^i, & t = a^i,\ x_i = a^i_i,\ x_j \neq a^i_j, \\ a^*, & t = a^i,\ x_i \neq a^i_i,\ x_j \neq a^i_j. \end{cases}$$

If exactly player i confirms, the next target is a^i. If both confirm, x binds.

The target is updated recursively from the public history, starting at a^*. Every target is reached as an intended agreement following some history, and all prescribed proposals are components of targets. Thus signalling of agreement holds.

I verify subgame perfection. At $x = t$, confirming gives $u_i(t)$, while rejecting gives

$$p\, u_i(t) + (1-p)\beta u_i(a^i).$$

Since every target gives player i at least $u_i(a^i) > 0$, confirming is strictly optimal.

Now suppose $x \neq t$, and let w be the target following two successful rejections. The gain to player i from rejecting rather than confirming, given that player j rejects, is

$$(1-p)\left\{\beta(1-p)[u_i(w) - u_i(a^i)] + p[\beta u_i(a^j) - u_i(x)]\right\}.$$

Whenever $w \neq a^i$, the first difference is strictly positive by conditions 2–4, so rejection is optimal for sufficiently small p.

The remaining case is $w = a^i$ for the punished player i. Here only player i changed its first-round proposal, so $x_j = a^i_j$. Condition 1 gives

$$u_i(x) \leq u_i(a^i) < \beta u_i(a^j),$$

and hence rejection is again optimal. The punisher's first difference is strictly positive by condition 4.

Finally, consider a unilateral first-round deviation by player i from a target t. At $p = 0$, both players subsequently reject, and the continuation target is

$$w_i(t) = \begin{cases} b^j, & t = a^j, \quad j \neq i, \\ t, & \text{otherwise.} \end{cases}$$

The deviation therefore gives $\beta u_i(w_i(t))$.

If $w_i(t) = t$, this is strictly below $u_i(t)$ since $u_i(t) > 0$ and $\beta < 1$. Otherwise $t = a^j$, and condition 4 gives

$$u_i(b^j) < u_i(a^j) = u_i(t).$$

Thus the deviation is strictly unprofitable at $p = 0$.

For $p > 0$, mistakes change the deviation payoff by at most $O(p)$, uniformly over proposals, since payoffs are bounded. The same first-round incentives therefore hold for sufficiently small p. The second-round comparisons also hold uniformly, using the fixed strict payoff gaps among the targets and the best-response inequality above.

Hence a single sufficiently small tremble probability rules out every profitable one-shot deviation. By continuity at infinity, the strategy is subgame perfect and supports a^*. ■

Proof of Theorem 5. Fix an equilibrium s^* and a history h at the beginning of a period. For any $\epsilon > 0$, take

$$\phi_i : A^*_{-i}(s^*) \to A_i$$

such that

$$u_i(\phi_i(a_{-i}), a_{-i}) \geq \sup_{a_i \in A_i} u_i(a_i, a_{-i}) - \epsilon$$

for every $a_{-i} \in A^*_{-i}(s^*)$.

Consider the deviation in which player i proposes $\phi_i(s^*_{-i}(h'))$ in the first round at every history h', and repeats that proposal in the second. By good faith, the opponent proposes an agreement action, so every resulting first-round profile pays player i at least $F^*_i - \epsilon$.

The opponent either confirms deliberately or confirms by mistake with probability p. Agreement therefore occurs in each period with conditional probability at least p.

For $F^*_i > 0$, take $\epsilon < F^*_i$. The deviation's payoff is at least

$$\begin{aligned} U_i(s'_i, s^*_{-i}|h) &\geq (F^*_i - \epsilon) \sum_{t=0}^{\infty} (1-p)^t p \beta^t \\ &= \varphi(F^*_i - \epsilon). \end{aligned}$$

Letting $\epsilon \to 0$ gives the required bound. When $F^*_i = 0$, the bound follows from non-negative payoffs. It is strictly positive whenever $F^*_i > 0$. ■

Cournot under discounting.

Take $u_i(q) = \pi_i(q) + bc_i$, and suppose

$$d = c_1 - c_2 > 0, \qquad k = b + c_2 - 2c_1 > 0.$$

Recall that the undiscounted punishment profiles are

$$\underline{q}^1 = (0, d + k), \qquad \underline{q}^2 = (k, d),$$

with punished-firm profits 0 and d^2, respectively.

For $\tau > 0$, consider the following punishments and intermediate agreements:

$$\underline{q}^1(\tau) = \left(\frac{\tau}{4}, d + k - \frac{\tau}{2}\right), \quad b^1(\tau) = \left(k - \frac{5\tau}{4}, d + \frac{\tau}{2}\right),$$

$$\underline{q}^2(\tau) = \left(k - \tau, d + \frac{\tau}{2}\right), \quad b^2(\tau) = \left(\frac{\tau}{4}, d + k - \frac{3\tau}{4}\right).$$

In each punishment, the punished firm best responds, so condition 1 of Proposition 2 holds. Their profits are

$$\pi_1(\underline{q}^1(\tau)) = \frac{\tau^2}{16}, \qquad \pi_2(\underline{q}^2(\tau)) = \left(d + \frac{\tau}{2}\right)^2,$$

which converge to the original punishment floors as $\tau \to 0$.

I now verify the intermediate-agreement conditions. For all sufficiently small $\tau > 0$,

$$\pi_1(\underline{q}^1) = \frac{\tau^2}{16} < \frac{\tau^2}{8} = \pi_1(b^2) < \pi_1(\underline{q}^2),$$
$$\pi_2(\underline{q}^2) = \left(d + \frac{\tau}{2}\right)^2 < \left(d + \frac{\tau}{2}\right)\left(d + \frac{3\tau}{4}\right) = \pi_2(b^1) < \pi_2(\underline{q}^1).$$

Moreover,

$$\pi_1(b^1) = \frac{3\tau}{4}\left(k - \frac{5\tau}{4}\right) > \pi_1(\underline{q}^1),$$
$$\pi_2(b^2) = \left(d + k - \frac{3\tau}{4}\right)\left(d + \frac{\tau}{2}\right) > \pi_2(\underline{q}^2).$$

Thus conditions 3 and 4 hold. The payoff normalisation does not affect these comparisons.

Finally, consider any Cournot profile q^* satisfying

$$\pi_1(q^*) > 0, \qquad \pi_2(q^*) > d^2.$$

Since the punishment payoffs converge to the original floors, condition 2 also holds for sufficiently small τ. The positivity and distinct-action requirements of the proposition are satisfied as well.

Proposition 2 therefore supports q^* whenever β is sufficiently close to one and p sufficiently small. Since the punishment payoffs converge to the original floors, the sufficient conditions hold for any agreement strictly above those floors, provided players are sufficiently patient and mistakes sufficiently rare. ■

Proofs for Section 6

Proof of lemma 2: By hypothesis there is no permissible coalition $C \in \mathcal{C}$ and $a'_C \in A_C$ with $u_i(a'_C, a^*_{-C}) > u_i(a^*)$ for all $i \in C$. A strong Nash equilibrium of G, and for a partition a coalitional equilibrium of Ray and Vohra (1997), satisfies this. Were a^* coalitionally irrational for some $C \in \mathcal{C}$, through a response ϕ_C, then evaluating ϕ_C at a^*_{-C} would furnish such an a'_C, a contradiction. Therefore a^* is not coalitionally irrational. Now suppose that $a^* \in \tilde{A}^m(\mathcal{C})$ for all $m < k$. Notice that by the same statement this implies that $a^* \in \tilde{A}^{m+1}(\mathcal{C})$. This implies that $a^* \in ICIR(\mathcal{C})$ for all $\mathcal{C}$. ■

Proof of theorem 6: Suppose not, for some history $h' \in H$ we have that $s_C(h') = a_C$. By coalitional signalling of agreement it follows that there exists some $h \in H$ such that $a_C(s|h) = a_C$. Therefore it must be that $U_i(s^*|h) = u_i(a(s^*|h)) \leq \sup_{a'_{-C} \in A_{-C}} u_i(a_C, a'_{-C})$

for all $i \in C$. By definition of a_C being not coalitionally rational, there exists a function $a'_C : A_{-C} \to A_C$ such that $\inf_{a_{-C} \in A_{-C}} u_i(a'_C(a_{-C}), a_{-C}) > \sup_{a'_{-C} \in A_{-C}} u_i(a_C, a'_{-C})$. Consider a deviation of C at history h such that $s_C(h') = a'_C(s_{-C}(h'))$ for all $h' \in H$. It follows that $U_i(s'_C, s^*_{-C}|h) \geq \inf_{a_{-C} \in A_{-C}} u_i(a'_C(a_{-C}), a_{-C}) > \sup_{a'_{-C} \in A_{-C}} u_i(a_C, a'_{-C}) \geq U_i(s^*|h)$ for all $i \in C$. Concluding that s^* is not a $\mathcal{C}$-Subgame Perfect Equilibrium.

Now suppose by contradiction that $s(h') \in \tilde{A}^k(\mathcal{C})\ \forall k < m$ and $h' \in H$ but $s(h') = a \notin \tilde{A}^{m+1}(\mathcal{C})$ for some $h' \in H$. By definition, it must be that $a \in \bigcup_{C \in \mathcal{C}} [D_C(\tilde{A}^{m-1}(\mathcal{C})_{-C}) \times A_{-C}]$. Therefore it must be that $a_C \in D_C(\tilde{A}^{m-1}(\mathcal{C})_{-C})$ for some $C \in \mathcal{C}$. By coalition agreement signalling we have that $\exists h \in H$ such that $a_C = a^*_C(s^*|h)$. By definition of coalition rationality given $\tilde{A}^{m-1}(\mathcal{C})_{-C}$, as $a_C \in D_C(\tilde{A}^{m-1}(\mathcal{C})_{-C})$ there must be some that there is some $a'_C : \tilde{A}^{m-1}(\mathcal{C})_{-C}$ such that $\inf_{a_{-C} \in \tilde{A}^{m-1}(\mathcal{C})_{-C}} u_i(a'_C(a_{-C}), a_{-C}) > \sup_{a'_{-C} \in \tilde{A}^{m-1}(\mathcal{C})_{-C}} u_i(a_C, a'_{-C})$. Consider a deviation of C at history h such that $s_C(h') = a'_C(s_{-C}(h'))$ for all $h' \in H$. It follows that:

$$U_i(s'_C, s^*_{-C}|h) \geq \inf_{a_{-C} \in \tilde{A}^{m-1}(\mathcal{C})_{-C}} u_i(a'_C(a_{-C}), a_{-C}) > \sup_{a'_{-C} \in \tilde{A}^{m-1}(\mathcal{C})_{-C}} u_i(a_C, a'_{-C})$$

Therefore $U_i(s'_C, s^*_{-C}|h) > U_i(s^*|h)$ for all $i \in C$. Concluding that s^* is not a $\mathcal{C}$-Subgame Perfect Equilibrium of the negotiation game. A contradiction. ■

Proof of theorem 7: Suppose this is not the case. There is some $C \in \mathcal{C}$ $a'_C : [ICIR(\mathcal{C})]_{-C} \to A_C$ such that $\inf_{a_{-C} \in [ICIR(\mathcal{C})]_{-C}} u_i(a'_C(a_{-C}), a_{-C}) > U_i(s^*|h)$ for all $i \in C$. It must be that s^* is a $\mathcal{C}$-Subgame Perfect Equilibrium of the negotiation game, and therefore there cannot exist a profitable deviation for C. Notice that $s^*_i(h) \in [ICIR(\mathcal{C})]_i$ for all $i \in N$.

Consider a joint deviation from coalition C such that $s'_C(h) = a'_C(s^*_{-C}(h))$ for all $h \in H$. By the definition of the utilities that this can induce, it is clear that: $U_i(s'_C, s^*_{-C}|h) \geq \inf_{a_{-C} \in [ICIR(\mathcal{C})]_{-C}} u_i(a'_C(a_{-C}), a_{-C})$ for all $i \in C$, and therefore $u_i(s'_C, s^*_{-C}|h) > U_i(s^*|h)$ for all $i \in C$. In conclusion, s^* cannot be a $\mathcal{C}$-Subgame Perfect Equilibrium.■

Proof of Theorem 8. Let $B = \{a^*\} \cup \{\underline{a}^C : C \in \mathcal{C}\}$. Since $\mathcal{C}$ is a partition, consider the auxiliary strategic-form game $\widehat{G}$ in which each coalition $C \in \mathcal{C}$ is a player, chooses $a_C \in A_C$, and receives

$$v_C(a) = \max_{i \in C} \{u_i(a) - m_i^C\}.$$

By definition of m_i^C, for every $b_C \in A_C$ we have $u_i(b_C, \underline{a}^C_{-C}) \leq m_i^C$ for each i, so $v_C(b_C, \underline{a}^C_{-C}) \leq 0$. The hypothesis gives $v_C(a) \geq 0$ for every $a \in B$, so in particular

$$v_C(\underline{a}^C) = \max_{b_C \in A_C} v_C(b_C, \underline{a}^C_{-C}) = 0.$$

Thus $\underline{a}^C_C$ is a best response to $\underline{a}^C_{-C}$ in $\widehat{G}$, and $v_C(a^*) \geq v_C(\underline{a}^C)$ and $v_C(\underline{a}^D) \geq v_C(\underline{a}^C)$ for ev-

ery $D \in \mathcal{C}$. These are the conditions of Theorem 2 for $\widehat{G}$. Apply its construction, taking the payoff of perpetual disagreement for the auxiliary coalition-player C to be $\limsup_t v_C(a^t)$. The resulting strategy profile reaches agreement after every history, supports a^*, and respects signalling of agreement.

Read this profile as a joint strategy for the members of each coalition in the original negotiation game. It respects coalitional signalling of agreement. Suppose some coalition C has a profitable joint deviation following a history h, where the prescribed continuation reaches an agreement $a \in B$. Take $i \in C$ with $v_C(a) = u_i(a) - m_i^C$. If the deviation reaches an agreement b, then $u_j(b) > u_j(a)$ for every $j \in C$, so

$$v_C(b) \geq u_i(b) - m_i^C > u_i(a) - m_i^C = v_C(a),$$

a profitable deviation for player C in $\widehat{G}$. If instead the deviation leads to perpetual disagreement with proposals $(a^t)_{t\geq 1}$, then the interval restriction and the assumed improvement of every member give $\limsup_t u_i(a^t) \geq U_i(s'_C, s^*_{-C} \mid h) > u_i(a)$, so

$$\limsup_t v_C(a^t) \geq \limsup_t \big(u_i(a^t) - m_i^C\big) > u_i(a) - m_i^C = v_C(a),$$

again a profitable deviation in $\widehat{G}$. Both contradict the equilibrium of $\widehat{G}$, so no permissible coalition improves all of its members. The lim sup above is only the valuation chosen for the auxiliary game. The argument uses only $\limsup_t u_i(a^t) \geq U_i$, which holds for every value the interval of Section 2 assigns to perpetual disagreement in the original game, so a^* is a $\mathcal{C}$-Negotiated Binding Agreement under each. ■

Proof of Remark 5. Let every player propose a^* at every history. The profile repeats and binds a^*, and the only action ever proposed is a component of a^*, so signalling of agreement holds. Suppose a permissible coalition C departs. The others go on proposing a^*_{-C}, so the profiles C can bring to agreement are those of the form (a^*_{-C}, a_C), none of which improves every member over a^* because a^* is a coalitional equilibrium. And should C never settle, each member's payoff is the $\liminf$ of $u_i(a^*_{-C}, a^t_C)$, which can exceed $u_i(a^*)$ for all i only if some late a^t_C beats a^* for every member, again ruled out. So no coalition can improve all its members, and a^* is a $\mathcal{C}$-Negotiated Binding Agreement. ■

B General Covers

The analysis in Section 6 has taken $\mathcal{C}$ to be a partition. The necessary conditions, Theorems 6 and 7, ask only that $\mathcal{C}$ cover N and hold for any cover. The richest cover is the full power set $2^N \setminus \{\emptyset\}$, under which every subset of players may deviate. The resulting

equilibrium is the strong perfect equilibrium of Rubinstein (1980), and I call its agreements *strong* Negotiated Binding Agreements. Three things change once coalitions overlap. First, the concept is only a stability requirement in the sense of Aumann (1959), the permissible coalitions being the groups entitled to deviate jointly and each tested on its own: it does not capture a player who uses membership in one coalition C to arrange, through a second coalition C', a deviation that benefits it at the expense of its partners in C, asking only for robustness against each coalition acting alone rather than modelling how coalitions form. Second, a coalition's deviation must be judged against all of its actions, not only the survivors, since an action deleted through one coalition a player belongs to may still belong to a profitable deviation of another coalition it belongs to. The invariance of the partition is then lost, as a coalition's guaranteeing deviation may itself be deleted by an overlapping coalition. Third, the deletion can lead to the empty set due to different coalitions containing the same player requiring different deletion, and then no agreement exists.

Corollary 2. *If $ICIR(\mathcal{C}) = \emptyset$ then no $\mathcal{C}$-Negotiated Binding Agreement can exist.*

This is immediate from Theorem 6, and the next example shows that this can be the case.

Example 2. Take the two-player game below, with $\mathcal{C} = \{\{1,2\},\{1\},\{2\}\}$. Only (M,R)

1\2	L	C	R
T	20,0	20,0	20,0
M	0,7.5	0,7.5	30,5
D	10,10	0,0	0,0

and (D,L) survive deletion for the grand coalition, and D is individually irrational for player 1, whose minmax is 20 while D pays at most 10, so after one round only (M,R) remains. But then R is coalitionally irrational for $\{2\}$ against the surviving M: it pays player 2 only 5, while against M player 2 guarantees 7.5 through the unrestricted deviation to L. Hence $ICIR(\mathcal{C}) = \emptyset$, and no agreement survives. ▼

Survival is not sufficient either, even where the deletion leaves a rich set behind.

Example 3. Let G be the two-player game below, with $\mathcal{C} = 2^N \setminus \{\emptyset\}$. The weakly Pareto

1\2	LL	L	R	RR
TT	**6,6**	0,4	**1,12**	0,0
T	4,0	0,0	**7,2**	1,1
D	**12,1**	**2,7**	4,4	0,8
DD	0,0	1,1	8,0	0,0

efficient profiles, in bold, are the only ones to survive the first round, and no further deletion

is possible, so

$$ICIR(2^N \setminus \{\emptyset\}) = \{(TT, LL), (TT, R), (T, R), (D, LL), (D, L)\}.$$

Yet (TT, R), though Pareto efficient and individually rational, paying player 1 its global minmax of 1, is not a strong Negotiated Binding Agreement: once the irrational actions are deleted player 1 guarantees 2 against the survivors $\{LL, L, R\}$, above what (TT, R) pays. Deletion can therefore rule out an outcome that is Pareto efficient and individually rational in G. ▼

Therefore allowing for a rich set of coalitions to deviate jointly may prevent existence at all, and therefore overcoming inefficiency is not a simple ask.

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